

Lectures on Groups and Lie Algebras

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Lecture Notes

Mathematics

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Chapter 1

Symmetries

In physics symmetries are often first introduced in the context of classical mechanics in order to simplify computations.

Definition 1.0.1: Symmetry

A **symmetry** is an invertible map

$$s : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

such that, whenever $x(t)$ is a solution of the equations of motion, also

$$s \circ x(t)$$

is a solution.

Example.

Consider a free particle, so that $\vec{F} = 0$. Then the equations of motion have solutions

$$x(t) = x_0 + tv_0, \quad x_0, v_0 \in \mathbb{R}^n.$$

For any $T \in \mathbb{R}^n$, the translation

$$s_T : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad x \mapsto x + T$$

is a symmetry.

Example.

In quantum mechanics, a symmetry is typically a unitary operator

$$S : \mathcal{H} \rightarrow \mathcal{H}$$

on the Hilbert space $\mathcal{H}$ such that

$$[S, H] = 0,$$

where $H : \mathcal{H} \rightarrow \mathcal{H}$ is the Hamiltonian.

For

$$\mathcal{H} = L^2(\mathbb{R}), \quad H = -\frac{1}{2m}\partial_x^2,$$

the translation operator

$$S_T : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R}), \quad (S_T f)(x) = f(x + T),$$

and the reflection operator

$$S_R : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R}), \quad (S_R f)(x) = f(-x),$$

are symmetries.

Abstract structure	Concrete realization
group G	representation ρ_V
$g \in G$	$\rho(g) : V \rightarrow V$
$e \in G$ (“do nothing”)	$\rho(e) = \text{id}_V$
$h \circ g \in G$ (symmetry composition)	$\rho(h \circ g) = \rho(h) \rho(g)$
$\exists g^{-1}$ s.t. $g^{-1} \circ g = g \circ g^{-1} = e$	$\rho(g^{-1}) \rho(g) = \rho(g) \rho(g^{-1}) = \text{id}_V$

Definition 1.0.2: Group

A **group** is a set G together with a map

$$\circ : G \times G \rightarrow G, \quad (g, h) \mapsto g \circ h,$$

such that:

1. $\circ$ is associative, that is,

$$(g \circ h) \circ k = g \circ (h \circ k) \quad \forall g, h, k \in G.$$

2. There exists an element $e \in G$ such that

$$e \circ g = g \circ e = g \quad \forall g \in G.$$

3. For every $g \in G$, there exists an element $g^{-1} \in G$ such that

$$g \circ g^{-1} = g^{-1} \circ g = e.$$

Corollary 1.0.3

In a group, the unit element and inverses are unique.

Proof:

Suppose e and e' are both unit elements. Then

$$e = e \circ e' = e'.$$

Hence the unit is unique.

Now let $g \in G$, and suppose h and h' are both inverses of g . Then

$$h = h \circ e = h \circ (g \circ h') = (h \circ g) \circ h' = e \circ h' = h'.$$

Hence the inverse of any element is unique. ■

Definition 1.0.4: Representation

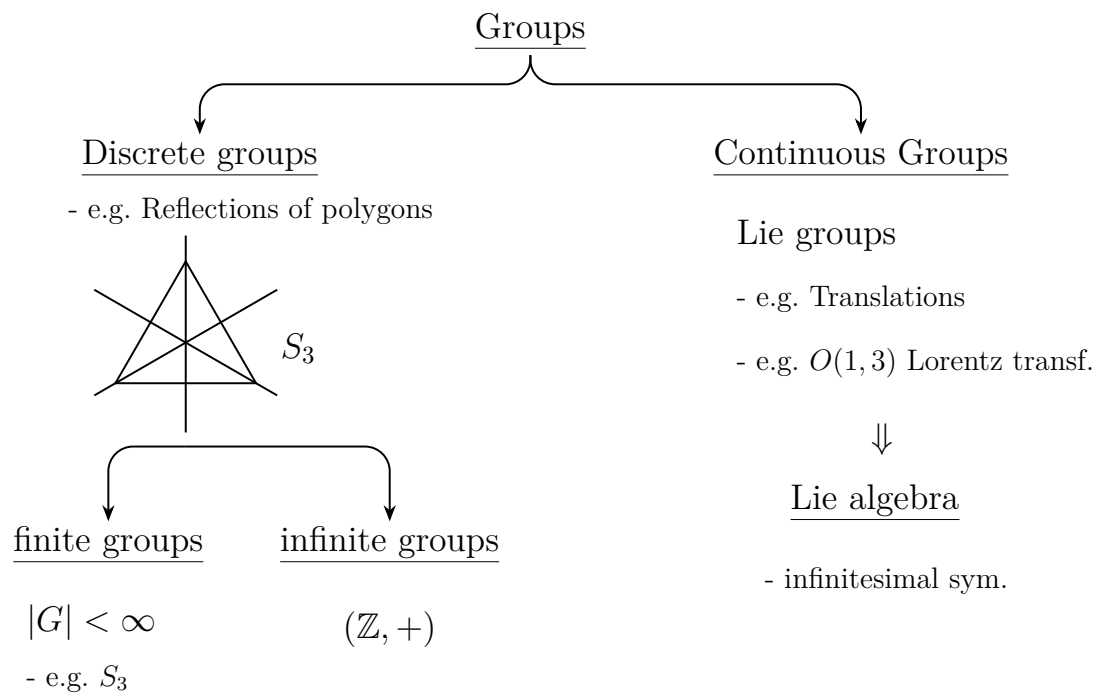
Let $(G, \circ)$ be a group. Usually we simply write G and leave the map $\circ$ implicit. A **representation** of G is a vector space V together with a map

$$\rho : G \rightarrow \text{End}(V)$$

such that:

1. $\rho(e) = \text{id}_V$.
2. $\rho(g)\rho(h) = \rho(g \circ h) \quad \forall g, h \in G$.

We will study these two definitions in this course. The basic structure of the course is as follows:



Chapter 2

Basic Group Theory

2.1 Homomorphisms And Isomorphisms

Definition 2.1.1: Group homomorphism

Let G and H be groups. A **group homomorphism** is a map

$$f : G \rightarrow H$$

such that

$$f(g_1 \circ_G g_2) = f(g_1) \circ_H f(g_2) \quad \forall g_1, g_2 \in G.$$

Corollary 2.1.2

If $f : G \rightarrow H$ is a group homomorphism, then

$$f(e_G) = e_H.$$

Corollary 2.1.3

If $f : G \rightarrow H$ and $\tilde{f} : H \rightarrow K$ are group homomorphisms, then

$$\tilde{f} \circ f : G \rightarrow K$$

is again a group homomorphism.

Corollary 2.1.4

If $f : G \rightarrow H$ is a bijective group homomorphism, then

$$f^{-1} : H \rightarrow G$$

is also a group homomorphism.

Corollary 2.1.5

Every group homomorphism preserves inverses, i.e. if $f : G \rightarrow H$ is a group homomorphism, then

$$f(g^{-1}) = f(g)^{-1} \quad \forall g \in G.$$

Proof:

Since $gg^{-1} = e_G$, we get

$$f(g)f(g^{-1}) = f(gg^{-1}) = f(e_G) = e_H.$$

Similarly,

$$f(g^{-1})f(g) = f(g^{-1}g) = f(e_G) = e_H.$$

Hence $f(g^{-1})$ is the inverse of $f(g)$, so $f(g^{-1}) = f(g)^{-1}$. ■

Definition 2.1.6: Group isomorphism

An invertible group homomorphism is called a **group isomorphism**. If there exists a group isomorphism

$$f : G \rightarrow H$$

then we say that G and H are **isomorphic**.

Example.

Examples of discrete groups:

- $(\mathbb{Z}, +)$.
- $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$.
- $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{C} \setminus \{0\}, \cdot)$.
- Let $n \in \mathbb{N}_{>0}$. Then $\mathbb{Z}/n\mathbb{Z}$ is the quotient of $\mathbb{Z}$ by the equivalence relation

$$a \sim b \quad \Longleftrightarrow \quad a = b + kn \quad \text{for some } k \in \mathbb{Z}.$$

The equivalence class of a is $[a] = \{b \in \mathbb{Z} \mid b \sim a\}$, and $\mathbb{Z}/n\mathbb{Z}$ becomes a group with operation

$$[a] \cdot [b] = [a + b].$$

- The permutation group S_n , for $n \in \mathbb{N}_{>0}$, is the group of all permutations of $\{1, 2, \dots, n\}$. With composition as multiplication, S_n forms a group, and $|S_n| = n!$.

For example, in S_3 let

$$\sigma_1 = \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array}, \quad \sigma_2 = \begin{array}{ccc} 1 & 2 & 3 \\ | & \times & \\ 1 & 2 & 3 \end{array}$$

Then

$$\sigma_2 \circ \sigma_1 = \begin{array}{ccc} 1 & 2 & 3 \\ & \diagdown & \diagup \\ & 1 & 2 \\ & \diagup & \diagdown \\ 1 & 2 & 3 \end{array}.$$

Example.

Examples of continuous groups:

- The general linear group

$$\mathrm{GL}(n, \mathbb{R}) = \{M : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid M \text{ linear, } \det M \neq 0\}.$$

- The orthogonal group

$$\mathrm{O}(n) = \{M : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid M \text{ linear, } M^T M = I\}.$$

Equivalently, $M \in \mathrm{O}(n)$ iff $\det M = \pm 1$.

- More generally, let $T_{ij} \in T\mathbb{R}^{n*} \otimes T\mathbb{R}^{n*}$ be a tensor. Its symmetry group is

$$\mathrm{Sym}(T_{ij}) = \{M^i_j \in \mathrm{GL}(n, \mathbb{R}) \mid T_{ij} = T_{ab} M^a_i M^b_j\}.$$

If $\langle v, w \rangle = T_{ij} v^i w^j$, this is equivalently the group preserving the bilinear form defined by T .

For

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

one gets $\mathrm{Sym}(T) = \mathrm{O}(1, 3)$. For the standard symplectic form

$$T = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

one gets the symplectic group.

- An **affine transformation** is a map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ of the form

$$x \mapsto Mx + T, \quad M \in \mathrm{GL}(n, \mathbb{R}), \quad T \in \mathbb{R}^n.$$

We denote the group of affine transformations by $\mathrm{Aff}(\mathbb{R}^n)$.

- If $B \subseteq \mathbb{R}^n$, define

$$\mathrm{Sym}(B) = \{A \in \mathrm{Aff}(\mathbb{R}^n) \mid A(B) = B\}.$$

For an equilateral triangle Δ , one has $\mathrm{Sym}(\Delta) \cong S_3$.

2.2 Subgroups

Definition 2.2.1: Subgroup

Let G be a group. A **subgroup** $H \leq G$ is a subset $H \subseteq G$ such that H is itself a group:

1. $h \circ_G h' \in H$ for all $h, h' \in H$,
2. if $h \in H$, then $h^{-1} \in H$.

Example.

Examples:

- $\text{SO}(n) \leq \text{O}(n) \leq \text{GL}(n)$.
- $S_2 \leq S_3 \leq S_4$.

This notation is slightly sloppy: strictly speaking, one should specify an injective group homomorphism $H \hookrightarrow G$. For example,

$$H = \{\sigma \in S_4 \mid \sigma(4) = 4\}$$

is isomorphic to S_3 .

Proposition 2.2.2

Let $f : G \rightarrow H$ be a group homomorphism. Then the kernel of f ,

$$\ker f := \{g \in G \mid f(g) = e_H\} \leq G,$$

is a subgroup. Likewise, the image of f ,

$$\text{im } f := \{h \in H \mid \exists g \in G, f(g) = h\} \leq H,$$

is a subgroup.

Proof:

Let $g_1, g_2 \in \ker f$. Then

$$f(g_1 \circ g_2) = f(g_1)f(g_2) = e_H,$$

so $g_1 \circ g_2 \in \ker f$. Also,

$$f(g^{-1}) = f(g)^{-1} = e_H$$

for every $g \in \ker f$, hence $g^{-1} \in \ker f$. Therefore $\ker f$ is a subgroup.

Now let $h_1, h_2 \in \text{im } f$. Then there exist $g_1, g_2 \in G$ such that $f(g_1) = h_1$ and $f(g_2) = h_2$. Hence

$$h_1 h_2 = f(g_1)f(g_2) = f(g_1 \circ g_2) \in \text{im } f.$$

Moreover, if $h = f(g) \in \text{im } f$, then

$$h^{-1} = f(g)^{-1} = f(g^{-1}) \in \text{im } f.$$

Thus $\text{im } f$ is a subgroup. ■

Example.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Then there is a group homomorphism

$$\text{Sym}(\Lambda) \rightarrow \text{O}(n), \quad f(x) = Ax + b \mapsto A.$$

The kernel is the group of translational symmetries of Λ , and the image is called the **point group**.

Example.

There is a group homomorphism

$$\text{sign} : S_n \rightarrow \{\pm 1\}.$$

It records whether a permutation is even or odd: even permutations are sent to 1, odd permutations to -1 .

For example,

$$\sigma = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ & \diagdown & \diagup & \\ & 1 & 2 & \\ & \diagup & \diagdown & \\ 1 & 2 & 3 & 4 \end{array} \Rightarrow \text{sign}(\sigma) = -1,$$

while

$$\tilde{\sigma} = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ & \diagdown & \diagup & \\ & 1 & 2 & \\ & \diagup & \diagdown & \\ 1 & 2 & 3 & 4 \end{array} \Rightarrow \text{sign}(\tilde{\sigma}) = 1.$$

Definition 2.2.3: Alternating group

The **alternating group** A_n is the kernel of the sign homomorphism:

$$A_n := \ker(\text{sign}) \leq S_n.$$

Equivalently, A_n is the subgroup of even permutations.

Example.

For $n = 3$,

$$A_3 = \left\{ \begin{array}{ccc} 1 & 2 & 3 \\ | & | & | \\ 1 & 2 & 3 \end{array}, \begin{array}{ccc} 1 & 2 & 3 \\ & \diagdown & \diagup \\ & 1 & 2 \\ & \diagup & \diagdown \\ 1 & 2 & 3 \end{array}, \begin{array}{ccc} 1 & 2 & 3 \\ & \diagup & \diagdown \\ & 1 & 2 \\ & \diagdown & \diagup \\ 1 & 2 & 3 \end{array} \right\}.$$

Hence A_3 has three elements and is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

2.3 Conjugacy Classes

Definition 2.3.1: Conjugation

Let G be a group and $g \in G$. Define the group homomorphism

$$\text{Ad}_g : G \rightarrow G, \quad h \mapsto ghg^{-1}.$$

This is called **conjugation by g** .

Indeed,

$$\text{Ad}_g(e) = geg^{-1} = e,$$

and

$$\text{Ad}_g(hh') = gh'h'g^{-1} = ghg^{-1}gh'g^{-1} = \text{Ad}_g(h)\text{Ad}_g(h').$$

Definition 2.3.2: Conjugate Elements

Let $h, h' \in G$. We say that h and h' are **conjugate** if there exists $g \in G$ such that

$$h = gh'g^{-1} = \text{Ad}_g(h').$$

Proposition 2.3.3

Being conjugate is an equivalence relation on G .

Proof:

- Reflexivity: every element is conjugate to itself, since $h = ehe^{-1}$.
- Symmetry: if $h \sim h'$, so $h = gh'g^{-1}$, then $h' = g^{-1}hg$, hence $h' \sim h$.
- Transitivity: if $h_1 \sim h_2$ and $h_2 \sim h_3$, say $h_1 = g_1h_2g_1^{-1}$ and $h_2 = g_2h_3g_2^{-1}$, then

$$h_1 = g_1g_2h_3(g_1g_2)^{-1},$$

so $h_1 \sim h_3$. ■

Definition 2.3.4: Conjugacy Classes

The equivalence classes of the conjugation relation are called **conjugacy classes**. The set of all conjugacy classes is denoted by $\text{conj}(G)$.

Example.

The conjugacy classes of S_3 are

$$\text{conj}(S_3) = \left\{ \{e\}, \left\{ \begin{array}{ccc} 1 & 2 & \\ & \times & \\ 1 & & 2 \end{array} \right\}, \left\{ \begin{array}{ccc} 3 & & \\ | & & \\ 3 & & \end{array} \right\}, \left\{ \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array} \right\}, \left\{ \begin{array}{ccc} 2 & 3 & \\ & \times & \\ 2 & & 3 \end{array} \right\} \right\},$$

$$\left\{ \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array} \right\}, \left\{ \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array} \right\} \right\}.$$

Remark.

A useful way to think about conjugacy classes is that conjugation changes the *description* of an object, but not the underlying structure.

For matrices, if $M \in \text{GL}(n, \mathbb{R})$, then

$$UMU^{-1}$$

is the same linear map written in a different basis.

For permutations, if $\sigma \in S_n$, then

$$\tau\sigma\tau^{-1}$$

simply relabels the elements of $\{1, \dots, n\}$. This is why the conjugacy classes in S_n are determined by cycle type.

Normal subgroups are exactly those subgroups that are stable under conjugation: if $N \trianglelefteq G$, then $gn g^{-1} \in N$ for all $g \in G$ and $n \in N$. This stability is what makes the quotient G/N well defined. This will be made precise in Section 2.4 .

Definition 2.3.5: Abelian Group

A group G is called **abelian** if

$$hg = gh \quad \forall h, g \in G.$$

Corollary 2.3.6

$$\text{conj}(G) = G \quad \Longleftrightarrow \quad G \text{ is abelian.}$$

Definition 2.3.7: Cycle

For $1 \leq k \leq n$, let $a_1, \dots, a_k \in \{1, 2, \dots, n\}$ be pairwise distinct. A **cycle** $(a_1, \dots, a_k)$ is the permutation

$$a_1 \mapsto a_2, \quad a_2 \mapsto a_3, \quad \dots, \quad a_k \mapsto a_1,$$

which acts as the identity on all other elements. We call k the **length** of the cycle.

Example.

To describe $\text{conj}(S_n)$, it is useful to use cycle type.

For $\sigma \in S_n$, the **cycle type** of σ is the list of lengths of the cycles in a disjoint cycle decomposition of σ , written in decreasing order. Thus the cycle type is a partition

$$\lambda_1 \geq \cdots \geq \lambda_k > 0, \quad \sum_{i=1}^k \lambda_i = n.$$

For example, $(1\ 3\ 2)(4\ 5) \in S_5$ has cycle type $(3, 2)$.

Theorem 2.3.8: Conjugacy In S_n

Two permutations $\sigma, \sigma' \in S_n$ are conjugate if and only if they have the same cycle type.

Proof:

Let $\tau, \sigma \in S_n$. Then

$$(\tau\sigma\tau^{-1})(\tau(i)) = \tau(\sigma(i)).$$

So conjugation by τ simply applies τ to the entries in each cycle. In particular,

$$\tau(a_1 \dots a_r)\tau^{-1} = (\tau(a_1) \dots \tau(a_r)).$$

Hence conjugate permutations have the same cycle type.

Conversely, if σ and σ' have the same cycle type, then their disjoint cycles can be matched cycle by cycle. Choosing τ so that it sends the entries of each cycle of σ to the corresponding entries of the matching cycle of σ' , we get

$$\tau\sigma\tau^{-1} = \sigma'.$$

Example.

The conjugacy classes of S_4 are in bijection with the partitions of 4:

$$(1, 1, 1, 1), \quad (2, 1, 1), \quad (3, 1), \quad (2, 2), \quad (4).$$

Representatives are:

- $(1, 1, 1, 1)$: e ,
- $(2, 1, 1)$: a transposition, e.g. $(1\ 2)$,
- $(3, 1)$: a 3-cycle, e.g. $(1\ 2\ 3)$,
- $(2, 2)$: a product of two disjoint transpositions, e.g. $(1\ 2)(3\ 4)$,
- (4) : a 4-cycle, e.g. $(1\ 2\ 3\ 4)$.

2.4 Normal Subgroups

Definition 2.4.1: Normal Subgroup

A subgroup $N \leq G$ is called **normal** if

$$gng^{-1} \in N \quad \forall n \in N, \forall g \in G.$$

Example.

In S_3 , the subgroup

$$\overline{N} = \{e, (12)\}$$

is not normal, since

$$(13)(12)(13)^{-1} = (23) \notin \overline{N}.$$

On the other hand,

$$N = \{e, (123), (132)\}$$

is a normal subgroup of S_3 .

Theorem 2.4.2: Quotient Group

Let $N \leq G$ be a normal subgroup. Let G/N be the set of equivalence classes for the relation $g \sim gn$ for $n \in N$. Then G/N is a group with multiplication

$$[g] \cdot [h] = [gh].$$

Proof:

We first check that the multiplication is well defined. If $g' = gn_1$ and $h' = hn_2$ with $n_1, n_2 \in N$, then

$$g'h' = gn_1hn_2 = gh(h^{-1}n_1h)n_2.$$

Since N is normal, $h^{-1}n_1h \in N$, hence $(h^{-1}n_1h)n_2 \in N$. Therefore $g'h' \sim gh$, so $[g'][h'] = [gh]$.

The group axioms are inherited from G : associativity follows from that in G , the neutral element is $[e]$, and the inverse of $[g]$ is $[g^{-1}]$.

The classes $[g] = gN \subseteq G$ are also called **cosets**. ■

Example.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Then the group of translational symmetries of Λ is normal, and

$$\text{Sym}(\Lambda)/\text{Trans}(\Lambda)$$

is the point group.

Another important example is $\mathbb{Z}^2 \leq \mathbb{R}^2$. Then $\mathbb{Z}^2$ is a normal subgroup of $(\mathbb{R}^2, +)$, and the quotient

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2$$

is the 2-torus.

Theorem 2.4.3: Isomorphism Theorem

Let $\varphi : G \rightarrow H$ be a group homomorphism.

1. The kernel $\ker \varphi$ is a normal subgroup of G .
2. The map

$$\Psi : G / \ker \varphi \rightarrow \operatorname{im} \varphi, \quad [g] \mapsto \varphi(g),$$

is a well-defined group isomorphism.

Proof:

Let $n \in \ker \varphi$ and $g \in G$. Then

$$\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g)e_H\varphi(g)^{-1} = e_H,$$

so $gng^{-1} \in \ker \varphi$. Hence $\ker \varphi \trianglelefteq G$.

Now define

$$\Psi([g]) := \varphi(g).$$

To see that Ψ is well defined, let $g' = gn$ with $n \in \ker \varphi$. Then

$$\varphi(g') = \varphi(gn) = \varphi(g)\varphi(n) = \varphi(g).$$

Also,

$$\Psi([g][h]) = \Psi([gh]) = \varphi(gh) = \varphi(g)\varphi(h) = \Psi([g])\Psi([h]),$$

so Ψ is a group homomorphism.

It is surjective by definition of $\operatorname{im} \varphi$. Finally, if $\Psi([g]) = \Psi([g'])$, then

$$\varphi(g) = \varphi(g') \implies \varphi(g^{-1}g') = e_H,$$

hence $g^{-1}g' \in \ker \varphi$, so $g' = g(g^{-1}g')$ and therefore $[g] = [g']$. Thus Ψ is injective. ■

Example.

Examples:

- If $f : G \rightarrow H$ is surjective, then $\operatorname{im} f = H$, so the isomorphism theorem gives

$$G / \ker f \cong H.$$

- Every finite abelian group A is a quotient of $\mathbb{Z}^n$ for some n . Hence

$$A \cong \mathbb{Z}^n / \ker f$$

for a suitable surjective homomorphism $f : \mathbb{Z}^n \rightarrow A$.

- Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. The homomorphism

$$\operatorname{Sym}(\Lambda) \rightarrow \operatorname{O}(n), \quad f(x) = Ax + b \mapsto A,$$

has kernel $\text{Trans}(\Lambda)$. Therefore

$$\text{Sym}(\Lambda)/\text{Trans}(\Lambda) \cong \text{im } f,$$

and $\text{im } f$ is the point group of Λ .

- The determinant defines a surjective homomorphism

$$\det : \text{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}_{>0},$$

where $\mathbb{R}_{>0}$ is viewed as a group under multiplication, and

$$\ker(\det) = \text{SL}(n, \mathbb{R}).$$

Hence

$$\text{GL}(n, \mathbb{R})/\text{SL}(n, \mathbb{R}) \cong \mathbb{R}_{>0}.$$

2.5 Extensions

Definition 2.5.1: Exact Sequence

A sequence of groups and group homomorphisms

$$\cdots \rightarrow G_i \xrightarrow{f_i} G_{i+1} \xrightarrow{f_{i+1}} G_{i+2} \rightarrow \cdots$$

is called **exact** if

$$\text{im}(f_i) = \ker(f_{i+1})$$

at every term.

Remark.

If $N \trianglelefteq G$, then the quotient map

$$\pi : G \rightarrow G/N, \quad g \mapsto [g],$$

fits into the exact sequence

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} G/N \rightarrow 1,$$

where $\iota : N \hookrightarrow G$ is the inclusion. Indeed, $\text{im}(\iota) = N = \ker(\pi)$.

By the isomorphism theorem, any short exact sequence

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$$

identifies N with a normal subgroup of G and H with the quotient G/N .

Definition 2.5.2: Extension

An **extension** of a group H by a group N is a short exact sequence

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1.$$

Definition 2.5.3: Split And Central Extensions

An extension

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$$

is called **split** if there exists a group homomorphism called a **section**

$$s : H \rightarrow G$$

such that

$$\pi \circ s = \text{id}_H.$$

It is called **central** if

$$\iota(n)g = g\iota(n) \quad \forall n \in N, \forall g \in G,$$

i.e. if $\iota(N)$ lies in the center $Z(G)$ of G .

Remark.

A split extension is the group-theoretic analogue of a trivial bundle. Indeed, the map $\pi : G \rightarrow H$ may be viewed as a bundle over H with fiber N , and a splitting $s : H \rightarrow G$ with $\pi \circ s = \text{id}_H$ is a global section. The existence of such a distinguished section means that the extension can be reconstructed from N and H without any twisting; in the simplest case one gets the direct product $N \times H$, and more generally a semidirect product.

Remark.

Central extensions are especially important in quantization. In classical mechanics one often starts with a symmetry group acting on the phase space, but after quantization the corresponding operators on the Hilbert space may realize the symmetry only up to phases. These extra phase factors are encoded by a central extension, typically by $U(1)$ or $\mathbb{R}$. In this sense, central extensions measure the obstruction to lifting a classical symmetry action to an honest linear action in quantum theory.

Example.

Examples:

- The product group $N \times H$ gives an extension

$$1 \rightarrow N \rightarrow N \times H \rightarrow H \rightarrow 1,$$

which is always split. If N is abelian, then this extension is also central.

- The symmetry group of a lattice fits into an extension

$$1 \rightarrow \text{Trans}(\Lambda) \rightarrow \text{Sym}(\Lambda) \rightarrow \text{point group} \rightarrow 1.$$

- The Lorentz group fits into the extension

$$1 \rightarrow \text{SO}^+(3, 1) \rightarrow \text{O}(3, 1) \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow 1,$$

which is split.

- The Poincare group P fits into the extension

$$1 \rightarrow \mathbb{R}^4 \rightarrow P \rightarrow \mathrm{O}(3, 1) \rightarrow 1,$$

which is split.

- The sequence

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 1$$

is a central extension, but it is not split.

Definition 2.5.4: Semidirect Product

Let $\psi : H \rightarrow \mathrm{Aut}(N)$ be a group homomorphism. The **semidirect product** $N \rtimes_{\psi} H$ is the group whose underlying set is $N \times H$, with multiplication

$$(n, h) \cdot (n', h') := (n \psi(h)[n'], hh').$$

In this notation one may think of the relation

$$hn = \psi(h)[n] h.$$

Remark.

There is a canonical split extension

$$1 \rightarrow N \xrightarrow{\iota} N \rtimes_{\psi} H \xrightarrow{\pi} H \rightarrow 1,$$

where

$$\iota(n) = (n, e_H), \quad \pi(n, h) = h, \quad s(h) = (e_N, h).$$

Thus every semidirect product gives a split extension, and conversely every split extension is isomorphic to a suitable semidirect product.

Theorem 2.5.5: Split Extensions And Semidirect Products

Let

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$$

be an extension, and let $s : H \rightarrow G$ be a splitting. Then:

1. Conjugation by $s(h)$ defines a group homomorphism

$$\psi : H \rightarrow \mathrm{Aut}(N), \quad h \mapsto (n \mapsto \iota^{-1}(s(h)\iota(n)s(h)^{-1})).$$

2. The group G is isomorphic to the semidirect product

$$N \rtimes_{\psi} H.$$

Proof:

Since $\iota(N) \trianglelefteq G$, conjugation by $s(h)$ preserves $\iota(N)$, hence defines an automorphism of N . This gives the map $\psi : H \rightarrow \text{Aut}(N)$, and it is a group homomorphism because conjugation is compatible with multiplication.

Define

$$\Phi : N \rtimes_{\psi} H \rightarrow G, \quad (n, h) \mapsto \iota(n)s(h).$$

Using the definition of ψ , one checks

$$\Phi((n, h)(n', h')) = \Phi(n, h)\Phi(n', h').$$

Thus Φ is a homomorphism.

It is surjective since every $g \in G$ can be written as $g = \iota(n)s(h)$, where $h = \pi(g)$ and n is determined by the fact that $gs(h)^{-1} \in \ker \pi = \iota(N)$. It is injective because $\iota(n)s(h) = e$ implies $h = e$ after applying π , and then $n = e$. Hence Φ is an isomorphism. ■

Remark.

If an extension is both split and central, then the associated action $\psi : H \rightarrow \text{Aut}(N)$ is trivial, because

$$\iota(n)s(h) = s(h)\iota(n) \quad \forall n \in N, h \in H.$$

Therefore

$$N \rtimes_{\psi} H = N \times H,$$

so every split central extension is in fact a direct product.

Chapter 3

Representation Theory

"How groups act on stuff."

3.1 Group Actions On Sets

Definition 3.1.1: Group Actions On Sets

Let G be a group and X a set.

A **left action** of G on X is a map

$$\triangleright : G \times X \rightarrow X, \quad (g, x) \mapsto g \triangleright x,$$

such that

$$h \triangleright (g \triangleright x) = (hg) \triangleright x, \quad e \triangleright x = x \quad \forall g, h \in G, \forall x \in X.$$

A **right action** of G on X is a map

$$\triangleleft : X \times G \rightarrow X, \quad (x, g) \mapsto x \triangleleft g,$$

such that

$$(x \triangleleft g) \triangleleft h = x \triangleleft (gh), \quad x \triangleleft e = x \quad \forall g, h \in G, \forall x \in X.$$

Equivalently, a left action is the same as a group homomorphism

$$\rho : G \rightarrow \text{Aut}(X), \quad g \mapsto (x \mapsto g \triangleright x),$$

and a right action is the same as an antihomomorphism

$$G \rightarrow \text{Aut}(X), \quad g \mapsto (x \mapsto x \triangleleft g).$$

Corollary 3.1.2

Left and right actions can be converted into each other using inverses. More precisely:

- if $\triangleright$ is a left action of G on X , then

$$x \triangleleft g := g^{-1} \triangleright x$$

defines a right action of G on X ;

- if $\triangleleft$ is a right action of G on X , then

$$g \triangleright x := x \triangleleft g^{-1}$$

defines a left action of G on X .

Proof:

Suppose $\triangleright$ is a left action and define $x \triangleleft g := g^{-1} \triangleright x$. Then

$$(x \triangleleft g) \triangleleft h = h^{-1} \triangleright (g^{-1} \triangleright x) = (h^{-1}g^{-1}) \triangleright x = (gh)^{-1} \triangleright x = x \triangleleft (gh),$$

and

$$x \triangleleft e = e^{-1} \triangleright x = x.$$

So this is a right action. The converse is proved in the same way. ■

Example.

Examples:

- Every group G acts trivially on any set X by

$$g \triangleright x = x \quad \forall g \in G, \forall x \in X.$$

- A group G acts on itself by conjugation:

$$g \triangleright h := ghg^{-1}.$$

Equivalently, this is the homomorphism

$$G \rightarrow \text{Aut}(G), \quad g \mapsto \text{Ad}_g.$$

- If $\Lambda \subseteq \mathbb{R}^n$ is a lattice, then its symmetry group $\text{Sym}(\Lambda)$ acts on Λ .
- The symmetric group S_n acts on the set $\{1, \dots, n\}$.
- The general linear group $\text{GL}(n, \mathbb{R})$ acts on $\mathbb{R}^n$.
- If $H \leq G$ is a subgroup, then G acts on the set of left cosets G/H by left multiplication:

$$g \triangleright [g'] := [gg'].$$

Definition 3.1.3: Orbit And Stabilizer

Let $\rho : G \rightarrow \text{Aut}(X)$ be an action and let $x \in X$.

The **orbit** of x , denoted by $\text{Orb}_G(x)$ or simply $\mathcal{O}_x$, is the set

$$\text{Orb}_G(x) = \{y \in X \mid g \triangleright x = y \text{ for some } g \in G\} \subseteq X.$$

The **stabilizer** of x , denoted by $\text{Stab}_G(x)$ or simply St_x , is the subgroup

$$\text{Stab}_G(x) = \{g \in G \mid g \triangleright x = x\} \leq G.$$

Example.

- Let G act on itself by conjugation. Then

$$\text{Orb}_G(g)$$

is the conjugacy class of g , and

$$\text{Stab}_G(g) = \{h \in G \mid hgh^{-1} = g\}$$

is the centralizer of g .

- The group $\text{SO}(2)$ acts on $\mathbb{R}^2$ by rotations. If $x \neq 0$, then

$$\text{Orb}_{\text{SO}(2)}(x) \cong S^1, \quad \text{Stab}_{\text{SO}(2)}(x) = \{e\}.$$

For the origin one has

$$\text{Orb}_{\text{SO}(2)}(0) = \{0\}, \quad \text{Stab}_{\text{SO}(2)}(0) = \text{SO}(2).$$

Theorem 3.1.4: Orbit-Stabilizer Theorem

Let $\rho : G \rightarrow \text{Aut}(X)$ be an action of a finite group G on a set X . Then for every $x \in X$,

$$|G| = |\text{Orb}_G(x)| |\text{Stab}_G(x)|.$$

Proof:

Write $St_x := \text{Stab}_G(x)$ and $\mathcal{O}_x := \text{Orb}_G(x)$. Since St_x need not be normal in G , the quotient G/St_x is considered only as the set of left cosets.

We claim that

$$\varphi : G/St_x \rightarrow \mathcal{O}_x, \quad [g] \mapsto g \triangleright x,$$

is a bijection.

First, φ is well-defined. If $[g'] = [g]$, then $g' = gh$ for some $h \in St_x$. Hence

$$\varphi([g']) = g' \triangleright x = (gh) \triangleright x = g \triangleright (h \triangleright x) = g \triangleright x = \varphi([g]).$$

The map is surjective by the definition of the orbit. To prove injectivity, suppose

$$g \triangleright x = g' \triangleright x.$$

Applying g^{-1} to both sides gives

$$(g^{-1}g') \triangleright x = x,$$

so $g^{-1}g' \in St_x$. Therefore $g' = g(g^{-1}g')$, and hence

$$[g'] = [g].$$

Thus φ is a bijection.

Consequently,

$$|G/St_x| = |\mathcal{O}_x|.$$

Since every left coset of St_x has $|St_x|$ elements,

$$|G| = |G/St_x| |St_x| = |\mathcal{O}_x| |St_x|.$$

Example.

- a) Let G be a finite group and let $H \leq G$. Then G acts on the set G/H of left cosets by left multiplication:

$$g \triangleright [g'] := [gg'].$$

The orbit of the coset $[e]$ is all of G/H , and its stabilizer is H . Indeed,

$$g \triangleright [e] = [e] \iff [g] = [e] \iff g \in H.$$

Hence the orbit-stabilizer theorem gives

$$|G| = |G/H| |H|.$$

In particular,

$$|H| \mid |G|.$$

Fix $g \in G$ and consider the cyclic subgroup

$$\langle g \rangle = \{g^k \mid k \in \mathbb{Z}\} \leq G.$$

Since G is finite, $\langle g \rangle$ is finite. Its size is called the **order** of g , denoted

$$\text{ord}(g) := |\langle g \rangle|.$$

Equivalently, $\text{ord}(g)$ is the smallest positive integer n such that $g^n = e$, and then

$$\langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}, \quad n = \text{ord}(g).$$

Applying the previous result to the subgroup $\langle g \rangle \leq G$ gives

$$\text{ord}(g) \mid |G|.$$

If $|G| = p$ is prime and $g \neq e$, then

$$\text{ord}(g) \neq 1 \quad \text{and} \quad \text{ord}(g) \mid p.$$

Therefore $\text{ord}(g) = p$, so

$$\langle g \rangle = G.$$

Thus every group of prime order is cyclic, and hence

$$G \cong \mathbb{Z}/p\mathbb{Z}.$$

b) What are the symmetries of a cube? Let

$$G = \text{Sym}([-1, 1]^3)$$

be the full symmetry group of the cube. Let F be the set of its faces. Then G acts on F . If $f \in F$ is one face, then the orbit of f is all of F , so

$$\mathcal{O}_f = F, \quad |\mathcal{O}_f| = 6.$$

The stabilizer of f is the symmetry group of a square face:

$$\text{Stab}_G(f) \cong D_4, \quad |D_4| = 8.$$

Hence the orbit-stabilizer theorem gives

$$|G| = |F| |D_4| = 6 \cdot 8 = 48.$$

Let $\text{Rot} \leq G$ be the subgroup of orientation preserving symmetries. The determinant gives a surjective homomorphism

$$\det : G \rightarrow \{\pm 1\} \cong \mathbb{Z}_2$$

with kernel Rot . Therefore

$$|G| = 2|\text{Rot}|, \quad |\text{Rot}| = 24.$$

There are four body diagonals in the cube. If D is the set of these diagonals, then rotations act on D , giving a homomorphism

$$\text{Rot} \rightarrow \text{Aut}(D) \cong S_4.$$

This action is faithful: a rotation fixing all four body diagonals must be the identity. Since both groups have order 24, we get

$$\text{Rot} \cong S_4.$$

Finally, the map

$$s : \mathbb{Z}_2 \rightarrow G, \quad -1 \mapsto -I_3$$

is a section of $\det$, and $-I_3$ commutes with every symmetry of the cube. Hence

$$G \cong S_4 \times \mathbb{Z}_2.$$

c) Let G act on itself by conjugation:

$$h \triangleright g := hgh^{-1}.$$

For $g \in G$, the orbit is the conjugacy class

$$\text{Orb}_G(g) = \{hgh^{-1} \mid h \in G\},$$

and the stabilizer is the centralizer

$$\text{Stab}_G(g) = \{h \in G \mid hgh^{-1} = g\} = \{h \in G \mid hg = gh\}.$$

Therefore the orbit-stabilizer theorem gives

$$|G| = |\{hgh^{-1} \mid h \in G\}| |\{h \in G \mid hg = gh\}|.$$

3.2 Representations On Vector Spaces

3.2.1 Representations And Intertwiners

Remark.

Recall that a **representation** of a group G on a vector space V is a group homomorphism

$$\rho : G \rightarrow \text{Aut}(V).$$

Thus each $g \in G$ acts on V by a structure-preserving map, namely an invertible linear map.

Definition 3.2.1: Maps Of Representations

Let (V, ρ) and (V', ρ') be representations of a group G .

A **map of representations**, or **intertwiner**, is a linear map

$$f : V \rightarrow V'$$

such that for every $g \in G$, the diagram

$$\begin{array}{ccc} V & \xrightarrow{f} & V' \\ \rho(g) \downarrow & & \downarrow \rho'(g) \\ V & \xrightarrow{f} & V' \end{array}$$

commutes. Equivalently,

$$f \circ \rho(g) = \rho'(g) \circ f \quad \forall g \in G.$$

Example.

- Let V be a vector space. Every group G has the **trivial representation** on V , defined by

$$\rho(g) = 1_V \quad \forall g \in G.$$

- The group $\mathbb{Z}_2 = \{[0], [1]\}$ acts on any vector space V by

$$\rho([0]) = 1_V, \quad \rho([1]) = -1_V.$$

- The orthogonal group $O(n)$ acts on $\mathbb{R}^n$ by its defining representation

$$\rho(A)(v) := Av.$$

- The symmetric group S_n acts on $\mathbb{R}$ through the sign homomorphism:

$$S_n \xrightarrow{\text{sgn}} \mathbb{Z}_2 \rightarrow \text{Aut}(\mathbb{R}), \quad \sigma \mapsto (x \mapsto \text{sgn}(\sigma)x).$$

- The symmetric group S_n acts on $\mathbb{R}^n$ by permuting the standard basis:

$$\rho(\sigma)(e_i) := e_{\sigma(i)}.$$

- Let X be a set equipped with a G -action. The free vector space

$$\mathbb{K}[X] := \text{span}_{\mathbb{K}}\{x \mid x \in X\}$$

carries a representation of G by linear extension:

$$\rho(g) \left(\sum_i \lambda_i x_i \right) := \sum_i \lambda_i (g \triangleright x_i).$$

- Taking $X = G$ with the left multiplication action gives the **regular representation** of G on $\mathbb{K}[G]$:

$$\rho(g)(h) := gh.$$

- The symmetry group of an equilateral triangle,

$$\text{Sym}(\triangle) \cong S_3 \cong D_3,$$

acts on $\mathbb{R}^2$. For example, if (12) is a reflection and (123) is a rotation, then one can choose coordinates such that

$$\rho((12)) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho((123)) = \begin{pmatrix} \cos(2\pi/3) & -\sin(2\pi/3) \\ \sin(2\pi/3) & \cos(2\pi/3) \end{pmatrix}.$$

- If $\Lambda \subseteq \mathbb{R}^n$ is a lattice, then its point group acts on $\mathbb{R}^n$.

3.2.2 Irreducibility And Schur's Lemma

Definition 3.2.2: Subrepresentations And Irreducibility

Let (V, ρ) be a representation of a group G .

A **subrepresentation** is a linear subspace $W \leq V$ such that

$$\rho(g)(w) \in W \quad \forall g \in G, \forall w \in W.$$

Equivalently, W is a G -invariant subspace of V .

The representation V is **irreducible** if its only subrepresentations are

$$\{0\} \quad \text{and} \quad V.$$

Proposition 3.2.3

Let

$$f : (V, \rho) \rightarrow (V', \rho')$$

be a map of G -representations. Then

$$\ker(f) \leq V \quad \text{and} \quad \text{im}(f) \leq V'$$

are subrepresentations.

Proof:

Let $v \in \ker(f)$. Then for every $g \in G$,

$$f(\rho(g)(v)) = \rho'(g)(f(v)) = \rho'(g)(0) = 0.$$

Hence $\rho(g)(v) \in \ker(f)$, so $\ker(f)$ is a subrepresentation of V .

Now let $v' \in \text{im}(f)$. Then $v' = f(v)$ for some $v \in V$. For every $g \in G$,

$$\rho'(g)(v') = \rho'(g)(f(v)) = f(\rho(g)(v)).$$

Thus $\rho'(g)(v') \in \text{im}(f)$, so $\text{im}(f)$ is a subrepresentation of V' . ■

Lemma 3.2.4: Schur's Lemma

Let (V, ρ) and (V', ρ') be irreducible representations of a group G . Then every map of G -representations

$$f : V \rightarrow V'$$

is either the zero map or an isomorphism.

Moreover, if V and V' are finite-dimensional complex vector spaces, then for any two isomorphisms of G -representations

$$f, f' : V \rightarrow V'$$

there exists $\lambda \in \mathbb{C}$ such that

$$f' = \lambda f.$$

Proof:

Let $f : V \rightarrow V'$ be a map of G -representations. By the previous proposition, $\ker(f)$ is a subrepresentation of V , and $\operatorname{im}(f)$ is a subrepresentation of V' . Since V and V' are irreducible, we have

$$\ker(f) \in \{\{0\}, V\}, \quad \operatorname{im}(f) \in \{\{0\}, V'\}.$$

If $\operatorname{im}(f) = \{0\}$, then $f = 0$. Otherwise $\operatorname{im}(f) = V'$. In this case $\ker(f) \neq V$, so $\ker(f) = \{0\}$. Hence f is both surjective and injective, and therefore an isomorphism.

Now assume that V and V' are finite-dimensional complex vector spaces, and let

$$f, f' : V \rightarrow V'$$

be isomorphisms of G -representations. Then

$$f^{-1} \circ f' : V \rightarrow V$$

is a map of G -representations. Since V is a finite-dimensional complex vector space, $f^{-1} \circ f'$ has an eigenvalue $\lambda \in \mathbb{C}$. Thus

$$f^{-1} \circ f' - \lambda 1_V$$

is not injective. By the first part, it cannot be an isomorphism, so it must be the zero map. Therefore

$$f^{-1} \circ f' = \lambda 1_V,$$

and multiplying by f gives

$$f' = \lambda f.$$

Example.

The second part of Schur's lemma need not hold over $\mathbb{R}$. Consider the representation of $\mathbb{Z}_4$ on $\mathbb{R}^2$ by rotations:

$$\rho([n]) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^n.$$

Then both

$$1_{\mathbb{R}^2} \quad \text{and} \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

are maps of $\mathbb{Z}_4$ -representations. Indeed, both commute with the action of the generator $[1]$, and hence with the action of every element of $\mathbb{Z}_4$.

However, these two maps are not scalar multiples of each other over $\mathbb{R}$, since the rotation matrix is not of the form $\lambda 1_{\mathbb{R}^2}$ for any $\lambda \in \mathbb{R}$. Thus the complex-vector-space assumption in Schur's lemma is essential for the scalar uniqueness statement.

Remark.

From now on, unless stated otherwise, representations will be taken over complex vector spaces.

3.2.3 Central Elements And Abelian Groups

Corollary 3.2.5

Let (V, ρ) be a finite-dimensional irreducible representation of a group G . Then every element of the center

$$Z(G) := \{c \in G \mid cg = gc \text{ for all } g \in G\}$$

acts by a scalar. In other words, for every $c \in Z(G)$ there exists $\lambda_c \in \mathbb{C}$ such that

$$\rho(c) = \lambda_c 1_V.$$

Proof:

Let $c \in Z(G)$. Then for every $g \in G$,

$$\rho(c) \circ \rho(g) = \rho(cg) = \rho(gc) = \rho(g) \circ \rho(c).$$

Thus $\rho(c) : V \rightarrow V$ is a map of G -representations. Since V is a finite-dimensional complex vector space, $\rho(c)$ has an eigenvalue $\lambda_c \in \mathbb{C}$. Then

$$\rho(c) - \lambda_c 1_V$$

is again a map of G -representations, but it is not injective. By Schur's lemma it cannot be an isomorphism, so it must be the zero map. Hence

$$\rho(c) = \lambda_c 1_V.$$

Example.

Let G be an abelian group. Then

$$Z(G) = G.$$

Hence, if (V, ρ) is a finite-dimensional irreducible representation of G , every element of G

acts by a scalar.

Therefore every linear subspace of V is preserved by the action of G . Since V is irreducible, this is only possible when

$$\dim V = 1.$$

Thus such representations are one-dimensional. Equivalently, they are homomorphisms

$$\rho : G \rightarrow \mathbb{C}^\times.$$

For example, let $G = \mathbb{Z}_n$. If $[1]$ is the generator, then

$$\rho([1])^n = \rho([0]) = 1.$$

Hence $\rho([1])$ must be an n -th root of unity. For $k \in \{0, 1, \dots, n-1\}$, define

$$\rho_k([1]) = e^{2\pi i k/n}.$$

Then

$$\rho_k([a]) = e^{2\pi i k a/n}$$

defines a one-dimensional irreducible representation of $\mathbb{Z}_n$, and these are all of them.

3.2.4 Decomposition Of Representations

Definition 3.2.6: Constructions From Old Representations

Let G be a group, and let (V_1, ρ_1) and (V_2, ρ_2) be representations of G .

- The **direct sum representation** is

$$(V_1 \oplus V_2, \rho_1 \oplus \rho_2),$$

where

$$(\rho_1 \oplus \rho_2)(g)(v_1, v_2) := (\rho_1(g)v_1, \rho_2(g)v_2).$$

- The **tensor product representation** is

$$(V_1 \otimes V_2, \rho_1 \otimes \rho_2),$$

where on pure tensors

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) := \rho_1(g)v_1 \otimes \rho_2(g)v_2,$$

and then extended linearly.

- The **dual representation** of (V_1, ρ_1) is

$$(V_1^*, \rho_1^*), \quad V_1^* := \text{Hom}(V_1, \mathbb{C}),$$

where

$$(\rho_1^*(g)f)(v) := f(\rho_1(g^{-1})v).$$

- The **Hom representation** is the representation on $\text{Hom}(V_1, V_2)$ defined by

$$\rho(g) : \text{Hom}(V_1, V_2) \rightarrow \text{Hom}(V_1, V_2),$$

with

$$(\rho(g)F)(v) := \rho_2(g)(F(\rho_1(g^{-1})v)).$$

Proposition 3.2.7

The direct sum, tensor product, dual, and Hom constructions above define representations of G .

Proof:

For the direct sum, one has

$$(\rho_1 \oplus \rho_2)(e)(v_1, v_2) = (v_1, v_2),$$

and

$$(\rho_1 \oplus \rho_2)(gh)(v_1, v_2) = (\rho_1(g)\rho_1(h)v_1, \rho_2(g)\rho_2(h)v_2) = ((\rho_1 \oplus \rho_2)(g)(\rho_1 \oplus \rho_2)(h))(v_1, v_2).$$

For the tensor product, the identity element acts as the identity. Moreover, on pure tensors,

$$(\rho_1 \otimes \rho_2)(gh)(v_1 \otimes v_2) = \rho_1(g)\rho_1(h)v_1 \otimes \rho_2(g)\rho_2(h)v_2,$$

which is equal to

$$((\rho_1 \otimes \rho_2)(g)(\rho_1 \otimes \rho_2)(h))(v_1 \otimes v_2).$$

Since pure tensors span $V_1 \otimes V_2$, this proves the representation identity on the whole tensor product.

For the dual representation, let $f \in V_1^*$ and $v \in V_1$. Then

$$(\rho_1^*(e)f)(v) = f(\rho_1(e^{-1})v) = f(v),$$

and

$$(\rho_1^*(gh)f)(v) = f(\rho_1((gh)^{-1})v) = f(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

On the other hand,

$$(\rho_1^*(g)(\rho_1^*(h)f))(v) = (\rho_1^*(h)f)(\rho_1(g^{-1})v) = f(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

Hence $\rho_1^*(gh) = \rho_1^*(g)\rho_1^*(h)$.

Finally, for the Hom representation, the identity element acts as the identity. For $F \in \text{Hom}(V_1, V_2)$ and $v \in V_1$, we compute

$$(\rho(gh)F)(v) = \rho_2(gh)F(\rho_1((gh)^{-1})v) = \rho_2(g)\rho_2(h)F(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

Also,

$$(\rho(g)(\rho(h)F))(v) = \rho_2(g)\left((\rho(h)F)(\rho_1(g^{-1})v)\right) = \rho_2(g)\rho_2(h)F(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

Thus $\rho(gh) = \rho(g)\rho(h)$, so this is a representation. ■

Remark.

Let V_1 and V_2 be finite-dimensional representations of G . Then there is a natural linear isomorphism

$$\Phi : V_2 \otimes V_1^* \rightarrow \text{Hom}(V_1, V_2), \quad v_2 \otimes f \mapsto (v_1 \mapsto f(v_1)v_2).$$

This is an isomorphism of representations. Indeed, for $g \in G$, $v_1 \in V_1$, $v_2 \in V_2$, and $f \in V_1^*$, we have

$$\Phi((\rho_2 \otimes \rho_1^*)(g)(v_2 \otimes f))(v_1) = f(\rho_1(g^{-1})v_1)\rho_2(g)v_2.$$

On the other hand,

$$(\rho(g)\Phi(v_2 \otimes f))(v_1) = \rho_2(g)\left(\Phi(v_2 \otimes f)(\rho_1(g^{-1})v_1)\right) = f(\rho_1(g^{-1})v_1)\rho_2(g)v_2.$$

Hence Φ intertwines the two representations.

Definition 3.2.8: Indecomposable Representation

A representation (V, ρ) of a group G is **indecomposable** if it cannot be written as a direct sum of two nonzero subrepresentations. In other words, there do not exist nonzero subrepresentations $W_1, W_2 \leq V$ such that

$$V = W_1 \oplus W_2.$$

Remark.

Every irreducible representation is indecomposable: if $V = W_1 \oplus W_2$ with W_1, W_2 nonzero subrepresentations, then W_1 and W_2 are proper nonzero subrepresentations of V , so V is not irreducible.

The converse is not true in general: an indecomposable representation can still have nonzero proper subrepresentations.

Example.

- Every one-dimensional representation is indecomposable, since a one-dimensional vector space cannot be written as a direct sum of two nonzero subspaces.
- Consider the representation of $\mathbb{Z}$ on $\mathbb{C}^2$ defined by

$$\rho(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then the line

$$W = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} \subseteq \mathbb{C}^2$$

is a subrepresentation, so the representation is reducible.

However, it is indecomposable. Indeed, a decomposition into two one-dimensional subrepresentations would give a basis of $\mathbb{C}^2$ consisting of eigenvectors of $\rho(1)$. This would make $\rho(1)$ diagonalizable. But

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not diagonalizable; its only eigenspace is W . Therefore this representation is reducible but indecomposable.

Definition 3.2.9: Semisimple Representations

Let (V, ρ) be a representation of a group G . We say that V is **semisimple**, or **completely reducible**, if every subrepresentation $W \leq V$ has a complementary subrepresentation. That is, for every subrepresentation $W \leq V$, there exists a subrepresentation $W' \leq V$ such that

$$V = W \oplus W'.$$

If V is finite-dimensional, this is equivalent to saying that

$$V \cong \bigoplus_i W_i$$

with each W_i irreducible.

Example.

Recall that

$$\mathrm{SU}(2) = \{U \in \mathrm{Mat}(2 \times 2, \mathbb{C}) \mid U^\dagger U = 1, \det(U) = 1\}$$

has a defining two-dimensional representation $V_{1/2} = \mathbb{C}^2$. For this representation one has

$$V_{1/2} \otimes V_{1/2} \cong \mathrm{Sym}^2(V_{1/2}) \oplus \bigwedge^2 V_{1/2}$$

as representations.

More generally, the irreducible representations of $\mathrm{SU}(2)$ are indexed by

$$j \in \frac{1}{2}\mathbb{N}_0,$$

and are given by

$$V_j := \mathrm{Sym}^{2j}(V_{1/2}).$$

The tensor product decomposes according to the Clebsch-Gordan rule

$$V_k \otimes V_\ell \cong \bigoplus_{j=|k-\ell|}^{k+\ell} V_j,$$

where j increases in steps of 1. This is the representation theoretic form of addition of angular momentum in quantum mechanics.

We will come back to this example later when discussing Lie groups.

Remark.

We want to show that, for finite groups, irreducible and indecomposable representations are the same thing. Equivalently, every finite-dimensional representation of a finite group should split as a direct sum of irreducible representations.

The subtle point is that a subspace does not have a preferred complement in general. For Hilbert spaces, however, there is a natural candidate: the orthogonal complement. This motivates the following definition.

Definition 3.2.10: Unitary Representation

A **unitary representation** of a group G is a Hilbert space $\mathcal{H}$ together with a group homomorphism

$$\rho : G \rightarrow U(\mathcal{H}),$$

where $U(\mathcal{H})$ denotes the group of unitary automorphisms of $\mathcal{H}$.

Proposition 3.2.11

Let (V, ρ) be a finite-dimensional representation of a finite group G . Then there exists a positive Hermitian inner product on V such that ρ is unitary. In other words, every finite-dimensional representation of a finite group can be lifted to a unitary representation.

Proof:

Let

$$\langle -, - \rangle$$

be any positive Hermitian inner product on V . Define a new inner product by averaging over the group:

$$\langle v, w \rangle_G := \sum_{g \in G} \langle \rho(g)v, \rho(g)w \rangle.$$

Since G is finite, this sum is finite.

The form $\langle -, - \rangle_G$ is again Hermitian and positive. Indeed, each summand is Hermitian, and for $v \neq 0$ one has

$$\langle v, v \rangle_G = \sum_{g \in G} \langle \rho(g)v, \rho(g)v \rangle > 0,$$

because every $\rho(g)$ is invertible, so $\rho(g)v \neq 0$.

It remains to check that the representation is unitary with respect to this new inner product. Let $h \in G$. Then

$$\langle \rho(h)v, \rho(h)w \rangle_G = \sum_{g \in G} \langle \rho(g)\rho(h)v, \rho(g)\rho(h)w \rangle$$

and hence

$$\langle \rho(h)v, \rho(h)w \rangle_G = \sum_{g \in G} \langle \rho(gh)v, \rho(gh)w \rangle.$$

As g runs through G , so does gh . Therefore, after the change of variables $g' = gh$,

$$\langle \rho(h)v, \rho(h)w \rangle_G = \sum_{g' \in G} \langle \rho(g')v, \rho(g')w \rangle = \langle v, w \rangle_G.$$

Thus every $\rho(h)$ preserves $\langle -, - \rangle_G$, so ρ is unitary. ■

Theorem 3.2.12: Maschke's Theorem

Every finite-dimensional representation of a finite group is completely reducible.

Proof:

Let (V, ρ) be a finite-dimensional representation of a finite group G . By the previous proposition, we may choose a positive Hermitian inner product

$$\langle -, - \rangle_G$$

on V such that ρ is unitary.

Let $W \leq V$ be a subrepresentation. Define its orthogonal complement with respect to this inner product by

$$W^\perp := \{v \in V \mid \langle v, w \rangle_G = 0 \text{ for all } w \in W\}.$$

Since V is finite-dimensional, we have a direct sum decomposition of vector spaces

$$V = W \oplus W^\perp.$$

It remains to show that $W^\perp$ is again a subrepresentation. Let $v \in W^\perp$, $g \in G$, and $w \in W$. Since $\rho(g)$ is unitary,

$$\langle \rho(g)v, w \rangle_G = \langle \rho(g)v, \rho(g)\rho(g^{-1})w \rangle_G = \langle v, \rho(g^{-1})w \rangle_G.$$

Because W is a subrepresentation, $\rho(g^{-1})w \in W$. Since $v \in W^\perp$, this gives

$$\langle \rho(g)v, w \rangle_G = 0.$$

Hence $\rho(g)v \in W^\perp$, so $W^\perp$ is a subrepresentation.

Thus every subrepresentation $W \leq V$ has a complementary subrepresentation $W^\perp$. Therefore V is completely reducible. ■

Corollary 3.2.13

Let G be a finite group. A finite-dimensional representation V of G is indecomposable if and only if it is irreducible. Equivalently, every finite-dimensional representation of G can be written as

$$V \cong \bigoplus_i N_i V_i,$$

where each V_i is irreducible and $N_i \geq 0$ is the multiplicity of V_i in V . Here

$$N_i V_i := \underbrace{V_i \oplus \cdots \oplus V_i}_{N_i \text{ times}}.$$

Proof:

By Maschke's theorem, every finite-dimensional representation of G is completely reducible.

We first prove the direct sum decomposition. If $V = 0$, there is nothing to show. Otherwise, choose a nonzero subrepresentation $W \leq V$ of minimal dimension. Then W is irreducible: any nonzero proper subrepresentation of W would contradict the minimality of W . By complete reducibility, there is a subrepresentation $W' \leq V$ such that

$$V = W \oplus W'.$$

Repeating this argument for W' , and using finite-dimensionality, the process terminates and gives a decomposition of V into irreducible subrepresentations. Grouping isomorphic irreducible summands gives

$$V \cong \bigoplus_i N_i V_i.$$

Now suppose that V is indecomposable. In such a decomposition, there can be only one irreducible summand and it can occur only once; otherwise V would split as a direct sum of two nonzero subrepresentations. Hence V is irreducible.

Conversely, every irreducible representation is indecomposable, as noted above. Therefore indecomposable and irreducible representations agree for finite groups. ■

Example.

Let

$$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$$

be the cyclic group, thought of as n equally spaced points on a circle. Consider the vector space of complex-valued functions

$$\mathcal{O}(\mathbb{Z}_n) := \{f : \mathbb{Z}_n \rightarrow \mathbb{C}\}.$$

The group $\mathbb{Z}_n$ acts on this space by translations:

$$(\rho([a])f)([x]) := f([x + a]).$$

For $k = 0, 1, \dots, n-1$, define

$$f_k([x]) := e^{2\pi i k x / n}.$$

These functions are eigenvectors for every translation, since

$$(\rho([a])f_k)([x]) = f_k([x + a]) = e^{2\pi i k(x+a)/n} = e^{2\pi i k a / n} f_k([x]).$$

Hence the line

$$\mathbb{C}_k := \mathbb{C}f_k$$

is a one-dimensional subrepresentation of $\mathcal{O}(\mathbb{Z}_n)$. Since the functions $f_0, \dots, f_{n-1}$ form a basis of $\mathcal{O}(\mathbb{Z}_n)$, we get the decomposition

$$\mathcal{O}(\mathbb{Z}_n) = \bigoplus_{k=0}^{n-1} \mathbb{C}_k = \bigoplus_{k=0}^{n-1} \mathbb{C}f_k.$$

Thus the decomposition of the regular representation of $\mathbb{Z}_n$ is exactly the discrete Fourier transform.

3.2.5 Character Theory

Definition 3.2.14: Characters And Class Functions

Let (V, ρ) be a finite-dimensional representation of a group G . The **character** of V is the function

$$\chi_V : G \rightarrow \mathbb{C}, \quad g \mapsto \text{tr}_V(\rho(g)).$$

A function $f : G \rightarrow \mathbb{C}$ is called a **class function** if it is invariant under conjugation, that is,

$$f(hgh^{-1}) = f(g) \quad \forall g, h \in G.$$

Proposition 3.2.15

Characters are class functions.

Proof:

Let (V, ρ) be a finite-dimensional representation of G . Then for every $g, h \in G$,

$$\chi_V(hgh^{-1}) = \text{tr}_V(\rho(hgh^{-1})) = \text{tr}_V(\rho(h)\rho(g)\rho(h^{-1})).$$

Using cyclicity of the trace, this becomes

$$\text{tr}_V(\rho(h)\rho(g)\rho(h^{-1})) = \text{tr}_V(\rho(g)\rho(h^{-1})\rho(h)) = \text{tr}_V(\rho(g)).$$

Hence

$$\chi_V(hgh^{-1}) = \chi_V(g),$$

so χ_V is invariant under conjugation. ■

Remark.

Since characters are class functions, it is enough to record their values on the conjugacy classes of G . A table whose rows are irreducible characters and whose columns are conjugacy classes is called a **character table**.

Example.

The group S_3 has three conjugacy classes, represented by

$$1, \quad (12), \quad (123).$$

It has three irreducible representations: the trivial representation, the alternating representation, and the standard representation. Their character table is

S_3	1	(12)	(123)
χ_{triv}	1	1	1
χ_{alt}	1	-1	1
χ_{std}	2	0	-1

Example.

For $\mathbb{Z}_3$, every conjugacy class consists of a single element. The irreducible characters are indexed by $k = 0, 1, 2$. Their character table is

$\mathbb{Z}_3$	[0]	[1]	[2]
$k = 0$	1	1	1
$k = 1$	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
$k = 2$	1	$e^{4\pi i/3}$	$e^{8\pi i/3}$

Remark.

Characters behave well with respect to direct sums and tensor products:

$$\chi_{V \oplus W} = \chi_V + \chi_W, \quad \chi_{V \otimes W} = \chi_V \chi_W.$$

The second identity follows from the fact that the trace of a tensor product of linear maps is the product of the traces.

Example.

For the standard representation of S_3 , the character is

$$\chi_{\text{std}} = (2, 0, -1)$$

on the conjugacy classes represented by $1, (12), (123)$. Therefore

$$\chi_{\text{std} \otimes \text{std}} = \chi_{\text{std}} \chi_{\text{std}} = (4, 0, 1).$$

On the other hand,

$$\chi_{\text{std}} + \chi_{\text{alt}} + \chi_{\text{triv}} = (2, 0, -1) + (1, -1, 1) + (1, 1, 1) = (4, 0, 1).$$

Hence the character table suggests the decomposition

$$\text{Std} \otimes \text{Std} \cong \text{Std} \oplus \text{Alt} \oplus \text{Triv}.$$

In particular, the characters in the table are linearly independent.

Definition 3.2.16: Invariant Subspace

Let G be a group and let (V, ρ) be a representation of G . The **subrepresentation of G -invariants** is

$$V^G := \{v \in V \mid \rho(g)(v) = v \text{ for all } g \in G\} \leq V.$$

Remark.

The subspace V^G is the part of V on which G acts trivially. Thus, for finite-dimensional representations of finite groups, one can write

$$V \cong \dim(V^G) \text{Triv} \oplus V',$$

where V' contains no copy of the trivial representation.

In particular, if V_i is irreducible but not trivial, then

$$V_i^G = \{0\}.$$

Proposition 3.2.17

Let G be a finite group and let (V, ρ) be a representation of G . The map

$$N_G : V \rightarrow V, \quad v \mapsto \frac{1}{|G|} \sum_{g \in G} \rho(g)(v)$$

is a projection operator onto V^G . In other words,

$$N_G \circ N_G = N_G, \quad \text{im}(N_G) = V^G.$$

Proof:

First we check that N_G intertwines the G -action. Let $g \in G$ and $v \in V$. Then

$$N_G(\rho(g)(v)) = \frac{1}{|G|} \sum_{h \in G} \rho(hg)(v).$$

As h runs through G , so does $h' = hg$. Therefore

$$N_G(\rho(g)(v)) = \frac{1}{|G|} \sum_{h' \in G} \rho(h')(v) = N_G(v).$$

Also,

$$\rho(g)(N_G(v)) = \rho(g) \left(\frac{1}{|G|} \sum_{h \in G} \rho(h)(v) \right) = \frac{1}{|G|} \sum_{h \in G} \rho(gh)(v).$$

Again, as h runs through G , so does $h' = gh$, and hence

$$\rho(g)(N_G(v)) = N_G(v).$$

Thus $\text{im}(N_G) \subseteq V^G$.

Conversely, if $v \in V^G$, then $\rho(g)(v) = v$ for every $g \in G$. Hence

$$N_G(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g)(v) = \frac{1}{|G|} \sum_{g \in G} v = v.$$

Therefore $V^G \subseteq \text{im}(N_G)$, and so

$$\text{im}(N_G) = V^G.$$

Finally, since $N_G(v) \in V^G$ for every $v \in V$, the previous computation gives

$$N_G(N_G(v)) = N_G(v).$$

Thus $N_G^2 = N_G$, so N_G is a projection operator onto V^G . ■

Corollary 3.2.18

Let G be a finite group and let V be a finite-dimensional representation of G . Then

$$\dim(V^G) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g).$$

In particular, if V is irreducible and not trivial, then

$$\sum_{g \in G} \chi_V(g) = 0.$$

Proof:

By the previous proposition, N_G is a projection onto V^G . Thus, with respect to a decomposition

$$V \cong V^G \oplus \ker(N_G),$$

the operator N_G has matrix

$$\begin{pmatrix} I_{\dim(V^G)} & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore

$$\text{tr}(N_G) = \dim(V^G).$$

On the other hand,

$$\text{tr}(N_G) = \text{tr} \left(\frac{1}{|G|} \sum_{g \in G} \rho(g) \right) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(\rho(g)) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g).$$

This proves the formula. If V is irreducible and not trivial, then $V^G = \{0\}$, and hence the sum of the character values is zero. ■

Remark.

If V and W are representations of G , then $\text{Hom}(V, W)$ is again a representation. Its invariant subspace is

$$\text{Hom}(V, W)^G = \{f : V \rightarrow W \mid f(\rho_V(g)v) = \rho_W(g)f(v) \text{ for all } g \in G, v \in V\}.$$

Thus $\text{Hom}(V, W)^G$ is precisely the space of maps of G -representations from V to W .

Corollary 3.2.19

Let V and W be irreducible finite-dimensional complex representations of a finite group G . Then

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Proof:

By Schur's lemma,

$$\mathrm{Hom}(V, W)^G = \begin{cases} \mathbb{C} & \text{if } V \cong W, \\ \{0\} & \text{otherwise.} \end{cases}$$

Hence

$$\dim \mathrm{Hom}(V, W)^G = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Applying the previous corollary to the representation $\mathrm{Hom}(V, W)$, we get

$$\dim \mathrm{Hom}(V, W)^G = \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}(V, W)}(g).$$

Since

$$\mathrm{Hom}(V, W) \cong V^* \otimes W,$$

its character is

$$\chi_{\mathrm{Hom}(V, W)}(g) = \chi_{V^*}(g) \chi_W(g) = \overline{\chi_V(g)} \chi_W(g).$$

Therefore

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Corollary 3.2.20

The characters of irreducible finite-dimensional complex representations of a finite group G are orthonormal with respect to the scalar product

$$(F, H) := \frac{1}{|G|} \sum_{g \in G} \overline{F(g)} H(g)$$

on the space of class functions on G .

Proof:

This is exactly the previous corollary, written in the notation

$$(\chi_V, \chi_W) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g).$$

Corollary 3.2.21

Every finite-dimensional representation V of a finite group G is completely determined, up to isomorphism, by its character χ_V . More precisely, if

$$V \cong \bigoplus_i N_i V_i$$

is a decomposition into irreducible representations, then

$$N_i = (\chi_{V_i}, \chi_V).$$

Proof:

By Maschke's theorem, write

$$V \cong \bigoplus_i N_i V_i$$

with the V_i irreducible. Then

$$\chi_V = \sum_i N_i \chi_{V_i}.$$

Taking the scalar product with χ_{V_j} , and using the orthonormality of irreducible characters, gives

$$(\chi_{V_j}, \chi_V) = \sum_i N_i (\chi_{V_j}, \chi_{V_i}) = N_j.$$

Thus the character determines all multiplicities N_i , and hence determines V up to isomorphism. ■

Remark.

Since irreducible characters are linearly independent, there can be only finitely many irreducible representations of a finite group up to isomorphism.

Example.

Let

$$D_4 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$$

be the symmetry group of the square, where r is rotation by 90° , and s is a reflection. Its conjugacy classes are

$$\{1\}, \quad \{r, r^3\}, \quad \{r^2\}, \quad \{s, sr^2\}, \quad \{sr, sr^3\}.$$

The one-dimensional representations are homomorphisms

$$\varphi : D_4 \rightarrow \mathbb{C}^\times.$$

Since

$$\varphi(r) = \varphi(srs) = \varphi(s)\varphi(r)\varphi(s) = \varphi(r)^{-1},$$

we must have $\varphi(r) = \pm 1$. Also, since $s^2 = 1$, one has $\varphi(s) = \pm 1$. Thus the one-dimensional

representations are labelled by the choices

$$(\varphi(r), \varphi(s)) \in \{(1, 1), (1, -1), (-1, 1), (-1, -1)\}.$$

There is also the standard two-dimensional representation

$$D_4 \rightarrow \mathrm{GL}(\mathbb{R}^2),$$

coming from the action of D_4 on the square in the plane. Its character is

$$\chi_{\mathrm{std}} = (2, 0, -2, 0, 0)$$

on the conjugacy classes listed above. Since

$$(\chi_{\mathrm{std}}, \chi_{\mathrm{std}}) = \frac{1}{8}(2^2 + (-2)^2) = 1,$$

the standard representation is irreducible.

The character table is

D_4	1	r	r^2	s	sr
$(1, 1)$	1	1	1	1	1
$(-1, 1)$	1	-1	1	1	-1
$(1, -1)$	1	1	1	-1	-1
$(-1, -1)$	1	-1	1	-1	1
std	2	0	-2	0	0

Here the column labels stand for the conjugacy classes represented by those elements.

Now consider $\mathrm{std} \otimes \mathrm{std}$. Its character is the pointwise product

$$\chi_{\mathrm{std} \otimes \mathrm{std}} = \chi_{\mathrm{std}} \chi_{\mathrm{std}} = (4, 0, 4, 0, 0).$$

The multiplicity of std in this tensor product is

$$(\chi_{\mathrm{std}}, \chi_{\mathrm{std} \otimes \mathrm{std}}) = \frac{1}{8}(2 \cdot 4 + (-2) \cdot 4) = 0.$$

For each one-dimensional character χ , the values on the classes represented by 1 and r^2 are both 1, so

$$(\chi, \chi_{\mathrm{std} \otimes \mathrm{std}}) = \frac{1}{8}(4 + 4) = 1.$$

Therefore

$$\mathrm{std} \otimes \mathrm{std} \cong (1, 1) \oplus (-1, 1) \oplus (1, -1) \oplus (-1, -1).$$

Corollary 3.2.22

Let G be a finite group, and let $V_1, \dots, V_r$ be the irreducible finite-dimensional complex representations of G , up to isomorphism. Then the regular representation decomposes as

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^r \dim(V_i) V_i.$$

In particular,

$$\sum_{i=1}^r \dim(V_i)^2 = |G|.$$

Proof:

Let χ_{reg} be the character of the regular representation $\mathbb{C}[G]$. In the basis $\{e_h\}_{h \in G}$, the action is

$$g \cdot e_h = e_{gh}.$$

If $g = e$, this is the identity on a vector space of dimension $|G|$, so

$$\chi_{\text{reg}}(e) = |G|.$$

If $g \neq e$, then $h \mapsto gh$ has no fixed points. Hence the matrix of g in the basis $\{e_h\}_{h \in G}$ is a permutation matrix with no diagonal entries equal to 1, and therefore

$$\chi_{\text{reg}}(g) = 0.$$

By the previous corollary, the multiplicity of V_i in $\mathbb{C}[G]$ is

$$(\chi_{V_i}, \chi_{\text{reg}}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} \chi_{\text{reg}}(g).$$

Since χ_{reg} vanishes away from id , this becomes

$$(\chi_{V_i}, \chi_{\text{reg}}) = \frac{1}{|G|} \overline{\chi_{V_i}(\text{id})} |G| = \dim(V_i).$$

Thus

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^r \dim(V_i) V_i.$$

Taking dimensions on both sides gives

$$|G| = \dim \mathbb{C}[G] = \sum_{i=1}^r \dim(V_i) \dim(V_i) = \sum_{i=1}^r \dim(V_i)^2.$$

Remark.

The previous corollary also gives a useful relation among the irreducible characters. Since

$$\chi_{\text{reg}} = \sum_{i=1}^r \dim(V_i) \chi_{V_i},$$

and since $\chi_{\text{reg}}(g) = 0$ for every $g \neq \text{id}$, one has

$$0 = \sum_{i=1}^r \dim(V_i) \chi_{V_i}(g) \quad \forall g \neq \text{id}.$$

Thus, once all irreducible characters except one are known, the remaining one is determined on all non-identity conjugacy classes by this equation. Its value at id is its dimension, which is constrained by

$$\sum_{i=1}^r \dim(V_i)^2 = |G|.$$

Theorem 3.2.23: Irreducible Characters Form A Basis

Let G be a finite group. The characters of the irreducible finite-dimensional complex representations of G form an orthonormal basis of the vector space of class functions on G .

In particular, the number of irreducible representations of G , up to isomorphism, is equal to the number of conjugacy classes of G .

Proof:

We already know that the irreducible characters are orthonormal. It remains to show that they span the space of class functions.

Let $\alpha : G \rightarrow \mathbb{C}$ be a class function such that

$$(\alpha, \chi_V) = 0$$

for every irreducible representation V . We will show that $\alpha = 0$.

For an irreducible representation (V, ρ) , define

$$\varphi_{\alpha, V} := \sum_{g \in G} \alpha(g) \rho(g) \in \text{End}(V).$$

Since α is a class function, $\varphi_{\alpha, V}$ commutes with the G -action. Indeed, for $h \in G$,

$$\varphi_{\alpha, V} \rho(h) = \sum_{g \in G} \alpha(g) \rho(gh).$$

Writing $g' = h^{-1}gh$, this becomes

$$\varphi_{\alpha, V} \rho(h) = \sum_{g' \in G} \alpha(hg'h^{-1}) \rho(hg') = \rho(h) \sum_{g' \in G} \alpha(g') \rho(g') = \rho(h) \varphi_{\alpha, V}.$$

Thus $\varphi_{\alpha, V}$ is a map of G -representations $V \rightarrow V$. By Schur's lemma,

$$\varphi_{\alpha, V} = \lambda_V 1_V$$

for some $\lambda_V \in \mathbb{C}$.

Taking traces gives

$$\lambda_V = \frac{1}{\dim(V)} \operatorname{tr}(\varphi_{\alpha,V}) = \frac{1}{\dim(V)} \sum_{g \in G} \alpha(g) \chi_V(g) = \frac{|G|}{\dim(V)} \overline{(\alpha, \chi_{V^*})} = 0.$$

Here V^* is again irreducible, so the last equality follows from the assumption on α . Hence $\varphi_{\alpha,V} = 0$ for every irreducible representation V .

Therefore, for any finite-dimensional representation (W, ρ_W) , one has

$$\sum_{g \in G} \alpha(g) \rho_W(g) = 0,$$

because W is a direct sum of irreducibles.

Apply this to the regular representation $\mathbb{C}[G]$. The operators $\rho_{\text{reg}}(g)$ are linearly independent: indeed,

$$\rho_{\text{reg}}(g)(e_{\text{id}}) = e_g.$$

Hence

$$\alpha(g) = 0 \quad \forall g \in G.$$

Thus $\alpha = 0$, and the irreducible characters span the class functions.

Since the irreducible characters form a basis, their number is the dimension of the space of class functions. A class function is determined by its values on conjugacy classes, so this dimension is the number of conjugacy classes of G . ■

3.3 Representations Of S_n

The irreducible representations of S_n are indexed by the conjugacy classes of S_n , and hence by the partitions of n .

Definition 3.3.1: Partitions And Young Diagrams

A **partition** of n is a tuple

$$\lambda = (\lambda_1, \dots, \lambda_k)$$

such that

$$n = \sum_{i=1}^k \lambda_i, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0.$$

We write $\lambda \vdash n$.

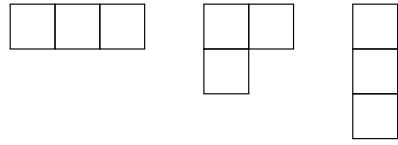
A partition can be represented by a **Young diagram**: draw λ_i boxes in the i -th row, aligned on the left.

Example.

The partitions of 3 are

$$(3), \quad (2, 1), \quad (1, 1, 1).$$

Their Young diagrams are



Thus S_3 has three irreducible representations, corresponding to these three diagrams.

Definition 3.3.2: Associative Algebras

An **associative algebra** $(A, \mu, 1)$ is a vector space A equipped with a linear map

$$\mu : A \otimes A \rightarrow A, \quad a \otimes b \mapsto ab,$$

and an element $1 \in A$ such that

$$(ab)c = a(bc) \quad \forall a, b, c \in A,$$

and

$$1a = a = a1 \quad \forall a \in A.$$

Example.

Let G be a group. The vector space $\mathbb{C}[G]$, with basis

$$\{e_g\}_{g \in G},$$

is an associative algebra with multiplication

$$e_g e_h = e_{gh},$$

and unit

$$1 = e_{\text{id}}.$$

It is called the **group algebra** of G .

Remark.

A representation

$$\rho : G \rightarrow \text{Aut}(V)$$

induces a linear map

$$\hat{\rho} : \mathbb{C}[G] \rightarrow \text{End}(V), \quad e_g \mapsto \rho(g).$$

In particular,

$$\hat{\rho} \left(\frac{1}{|G|} \sum_{g \in G} e_g \right)$$

is the projection onto the trivial component of V .

Definition 3.3.3: Tableaux

A **tableau** T on a given Young diagram with n boxes is a numbering of its boxes by $1, \dots, n$.

Associated to T are two subgroups of S_n :

$$P_T := \{\sigma \in S_n \mid \sigma \text{ preserves the rows of } T\},$$

and

$$Q_T := \{\sigma \in S_n \mid \sigma \text{ preserves the columns of } T\}.$$

Example.

For example, for the tableau

$$T = \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & \\ \hline 6 & 7 & \\ \hline \end{array}$$

one has

$$P_T \cong S_3 \times S_2 \times S_2, \quad Q_T \cong S_3 \times S_3 \times S_1.$$

We will suppress the tableau from the notation and write λ for both a partition and a choice of tableau of shape λ .

Definition 3.3.4: Young Symmetrizers

Let $\lambda \vdash n$, and choose a tableau of shape λ . Define

$$a_\lambda := \sum_{g \in P_\lambda} e_g \in \mathbb{C}[S_n],$$

and

$$b_\lambda := \sum_{g \in Q_\lambda} \text{sign}(g) e_g \in \mathbb{C}[S_n].$$

The **Young symmetrizer** associated to λ is

$$c_\lambda := a_\lambda b_\lambda \in \mathbb{C}[S_n].$$

Example.

For $\lambda = (n)$, the Young diagram has one row. Hence

$$P_\lambda = S_n, \quad Q_\lambda = \{\text{id}\},$$

and so

$$c_\lambda = a_\lambda.$$

In this case

$$\text{im}((\cdot)c_\lambda) \cong \text{Triv}.$$

For $\lambda = (1, \dots, 1)$, the Young diagram has one column. Hence

$$P_\lambda = \{\text{id}\}, \quad Q_\lambda = S_n,$$

and so

$$a_\lambda = 1, \quad c_\lambda = b_\lambda.$$

In this case

$$\text{im}((\cdot)c_\lambda) \cong \text{Alt}.$$

Definition 3.3.5: Representations From Young Symmetrizers

Let $\lambda \vdash n$. Define

$$V_\lambda := \mathbb{C}[S_n]c_\lambda = \{ac_\lambda \mid a \in \mathbb{C}[S_n]\} \leq \mathbb{C}[S_n].$$

This is a representation of S_n by left multiplication:

$$\rho(g) : V_\lambda \rightarrow V_\lambda, \quad ac_\lambda \mapsto e_g ac_\lambda.$$

Example.

For S_3 , the three partitions give

$$V_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}} = \mathbb{C}[e_{\text{id}} + e_{(12)} + e_{(23)} + e_{(13)} + e_{(123)} + e_{(132)}],$$

and

$$V_{\begin{array}{|c|} \hline \square \\ \square \\ \square \\ \hline \end{array}} = \mathbb{C}[e_{\text{id}} - e_{(12)} - e_{(23)} - e_{(13)} + e_{(123)} + e_{(132)}].$$

For the tableau

1	2
3	

one has

$$c_\lambda = (e_{\text{id}} + e_{(12)})(e_{\text{id}} - e_{(13)}) = e_{\text{id}} + e_{(12)} - e_{(13)} - e_{(123)}.$$

Hence

$$V_{\begin{array}{|c|c|} \hline \square & \square \\ \square & \square \\ \hline \end{array}} = \mathbb{C}[S_3]c_\lambda.$$

Proposition 3.3.6

Let T, T' be Young tableaux of the same shape λ . Then there exists a unique $g \in S_n$ such that

$$T' = gT.$$

If P, Q are the row and column subgroups associated to T , and P', Q' are the row and column subgroups associated to T' , then

$$P' = gPg^{-1}, \quad Q' = gQg^{-1}.$$

Hence

$$c_{\lambda, T'} = e_g c_{\lambda, T} e_{g^{-1}}.$$

Therefore

$$V_{\lambda, T} \rightarrow V_{\lambda, T'}, \quad ac_{\lambda, T} \mapsto ac_{\lambda, T} e_{g^{-1}} = ae_{g^{-1}} c_{\lambda, T'},$$

is an isomorphism of S_n -representations.

Moreover,

$$P \cap Q = \{\text{id}\}.$$

Thus, if

$$g = pq = p'q', \quad p, p' \in P, \quad q, q' \in Q,$$

then

$$p'^{-1}p = qq'^{-1} \in P \cap Q.$$

Hence $p = p'$ and $q = q'$, so the decomposition $g = pq$ is unique. Consequently,

$$c_{\lambda, T} = \sum_{g=pq} \text{sign}(q) e_g.$$

Proposition 3.3.7

For each Young tableau, the representation V_λ is irreducible.

Proof:

Let $v \in V_\lambda$. We want to understand $c_\lambda v$. Write

$$v = ac_\lambda.$$

Then $c_\lambda v = c_\lambda ac_\lambda$. Hence $c_\lambda v$ solves the equation

$$e_p x e_q \text{sign}(q) = x \quad \forall p \in P, q \in Q. \tag{1}$$

Write

$$x = \sum_{g \in S_n} n_g e_g.$$

Equation (1) implies

$$n_{pgq} = n_g \text{sign}(q).$$

If $g \neq pq$, then let $T' = gT$. Assume that there are integers $i < j \leq n$ such that i, j are in the same row of T and in the same column of T' . Then $t = (ij) \in P$, and also

$$t \in Q' = gQg^{-1}.$$

Thus there is some $q \in Q$ with $t = gqg^{-1}$, and hence

$$g = tqg^{-1}.$$

Comparing coefficients gives

$$n_g = -n_g, \quad \text{so } n_g = 0.$$

Now let $T, T' = gT$ be two Young tableaux such that no pair i, j is in the same row of T and in the same column of T' . Then there are $p \in P$ and $q' \in Q'$ such that the first rows of pT and $q'T'$ agree. Continuing row by row, we get

$$pT = q'T' = q'gT.$$

Hence

$$p = q'g.$$

Since $q' \in Q' = gQg^{-1}$, we can write $q' = gqg^{-1}$ for some $q \in Q$. Therefore

$$p = gq, \quad g = pq^{-1}.$$

It follows that the only possible non-zero coefficients are those with $g = pq$. Thus every solution of (1) is of the form

$$x = n_1 \sum_{g=pq} e_g \text{sign}(g) = n_1 c_\lambda.$$

Hence

$$c_\lambda V_\lambda \subseteq \mathbb{C}[c_\lambda].$$

Let $W \subseteq V_\lambda$ be a subrepresentation. Then either

$$c_\lambda W = 0 \quad \text{or} \quad c_\lambda W = \mathbb{C}[c_\lambda].$$

In the second case, $c_\lambda \in W$, and hence

$$ac_\lambda \in W \quad \forall a \in \mathbb{C}[S_n].$$

Thus $V_\lambda \subseteq W$, so $W = V_\lambda$.

In the first case, choose a projection operator

$$P_W : \mathbb{C}[S_n] \rightarrow \mathbb{C}[S_n]$$

which projects onto W . It satisfies

$$e_g P_W(a) = P_W(e_g a).$$

For $a = 1$, this gives

$$e_g P_W(1) = P_W(e_g).$$

Hence P_W is right multiplication by $P_W(1)$. Since $P_W(1) \in W$, the assumption $c_\lambda W = 0$ gives

$$P_W(c_\lambda) = c_\lambda P_W(1) = 0.$$

Therefore, for every $a \in \mathbb{C}[S_n]$,

$$P_W(ac_\lambda) = aP_W(c_\lambda) = 0.$$

Since $W \subseteq V_\lambda$ and P_W is the identity on W , we get $W = 0$.

Thus the only subrepresentations are 0 and V_λ . ■

Remark.

The proof shows that

$$c_\lambda V_\lambda \neq 0.$$

Theorem 3.3.8: Classification Of Irreducible Representations Of S_n

For $\lambda \neq \mu$, the representations V_λ and V_μ are not isomorphic.

Hence the V_λ , where $\lambda \vdash n$, form a complete set of irreducible representations of S_n .

Proof:

Assume $\lambda > \mu$, meaning that the first non-zero difference $\lambda_i - \mu_i$ is positive. We show that

$$c_\lambda V_\mu = \{0\}, \quad c_\lambda V_\lambda \neq 0.$$

The second statement was shown in the previous proof. For the first one, it is enough to show that

$$c_\lambda a c_\mu = 0 \quad \forall a \in \mathbb{C}[S_n].$$

Since

$$c_\lambda a c_\mu = a_\lambda b_\lambda a a_\mu b_\mu,$$

it is enough to prove

$$a_\lambda x b_\mu = 0 \quad \forall x \in \mathbb{C}[S_n].$$

It suffices to take $x = e_g$. Then

$$a_\lambda e_g b_\mu = a_\lambda b_{gT_\mu}.$$

Since $\lambda > \mu$, there exist i, j which lie in the same row of T_λ and in the same column of gT_μ .

Let $t = (ij)$. Then $t \in P_\lambda$, so

$$a_\lambda e_t = a_\lambda,$$

while t lies in the column group of gT_μ , so

$$e_t b_{gT_\mu} = -b_{gT_\mu}.$$

Therefore

$$a_\lambda b_{gT_\mu} = a_\lambda e_t e_t b_{gT_\mu} = -a_\lambda b_{gT_\mu}.$$

Hence

$$a_\lambda b_{gT_\mu} = 0.$$

Thus $c_\lambda V_\mu = \{0\}$.

Since $c_\lambda V_\lambda \neq 0$, the representations V_λ and V_μ cannot be isomorphic. We have already shown that every V_λ is irreducible, and there are as many partitions $\lambda \vdash n$ as conjugacy classes in S_n . Therefore these representations classify all irreducible representations of S_n . ■

For

$$i = (i_1, \dots, i_n), \quad \sum_{\alpha=1}^n i_\alpha \alpha = n,$$

we denote by C_i the conjugacy class of S_n consisting of permutations with i_α many α -cycles.

Let $\lambda \vdash n$ be a partition with k rows. We work with polynomials in k variables

$$x_1, \dots, x_k.$$

For $1 \leq j \leq n$, define

$$p_j(x) := x_1^j + \dots + x_k^j,$$

and define

$$\Delta(x) := \prod_{a < b} (x_a - x_b).$$

For a polynomial $Q(x)$ in $x_1, \dots, x_k$, we denote by

$$[Q(x)]_{\ell_1, \dots, \ell_k}$$

the coefficient of

$$x_1^{\ell_1} \dots x_k^{\ell_k}.$$

Here

$$\ell_1 = \lambda_1 + k - 1, \quad \ell_2 = \lambda_2 + k - 2, \quad \dots, \quad \ell_k = \lambda_k.$$

Theorem 3.3.9: Frobenius Formula

With the notation above,

$$\chi_\lambda(C_i) = \left[\Delta(x) \prod_{j=1}^n p_j(x)^{i_j} \right]_{\ell_1, \dots, \ell_k}.$$

Proof:

See (*FultonHarris*) ■

Example.

Let $n = 5$, and let

$$\lambda = (3, 2) = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \\ \hline \end{array}.$$

Consider

$$i = (0, 1, 1, 0, 0),$$

so that

$$(12)(345) \in C_i.$$

Here $k = 2$, hence

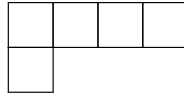
$$\ell_1 = \lambda_1 + 1 = 4, \quad \ell_2 = \lambda_2 = 2.$$

Therefore the Frobenius formula gives

$$\chi_\lambda(C_i) = \left[(x_1 - x_2)(x_1^2 + x_2^2)(x_1^3 + x_2^3) \right]_{4,2} = 1.$$

Remark.

The representation corresponding to



is the standard representation: S_n acts on $\mathbb{C}^n$ by permuting coordinates. This representation is not irreducible, since

$$e_1 + \cdots + e_n \in (\mathbb{C}^n)^{S_n}.$$

The subspace

$$V := \left\{ (x_1, \dots, x_n) \in \mathbb{C}^n \mid \sum_{i=1}^n x_i = 0 \right\}$$

is the standard S_n -representation.

More generally, the representation corresponding to

$$(n - k, 1, \dots, 1)$$

is

$$\bigwedge^k V.$$

If λ' denotes the conjugate partition, then

$$V_\lambda = V_{\lambda'} \otimes \text{Alt}.$$

3.4 Applications of Group Theory in Quantum Mechanics

3.4.1 Symmetries of the Hamiltonian

In quantum mechanics, the state space of a particle in $\mathbb{R}^n$ is the Hilbert space

$$\mathcal{H} = L^2(\mathbb{R}^n).$$

Elements of $\mathcal{H}$ are wave functions

$$\psi : \mathbb{R}^n \rightarrow \mathbb{C}$$

with inner product

$$\langle \psi | \varphi \rangle = \int \overline{\psi(x)} \varphi(x) dx.$$

Definition 3.4.1: Affine Group

The **group of affine transformations** of $\mathbb{R}^n$ is

$$\text{Aff}_n := \{\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid \alpha(x) = Ax + T, \ A \in \text{GL}(n, \mathbb{R}), \ T \in \mathbb{R}^n\}.$$

Identifying α with the pair (T, A) acting by

$$(T, A) : x \mapsto Ax + T,$$

composition reads

$$(T, A)(S, B) = (T + AS, AB),$$

and consequently

$$\text{Aff}_n \cong \mathbb{R}^n \rtimes \text{GL}(n, \mathbb{R}).$$

Remark.

There is a natural group homomorphism which remembers only the linear part:

$$\text{Aff}_n \rightarrow \text{GL}(n, \mathbb{R}), \quad (T, A) \mapsto A,$$

The kernel consists of the transformations with $A = \text{id}$, namely

$$x \mapsto x + T.$$

Hence the translations form a normal subgroup of Aff_n .

In quantum mechanics, one is interested in transformations that preserve distances, angles, and the Lebesgue measure. This restricts the linear part from $\text{GL}(n, \mathbb{R})$ to $O(n)$, and hence gives the subgroup

$$\mathbb{R}^n \rtimes O(n) = \{x \mapsto Ax + T \mid A \in O(n), \ T \in \mathbb{R}^n\}.$$

For $\alpha(x) = Ax + T$, one has

$$\alpha^{-1}(x) = A^{-1}(x - T).$$

The corresponding action on a wave function f is by pullback along the inverse:

$$\rho(\alpha)f(x) = f(\alpha^{-1}(x)) = (f \circ \alpha^{-1})(x).$$

For $a, b \in \text{Aff}_n$,

$$\rho(a)\rho(b)f(x) = (f \circ b^{-1} \circ a^{-1})(x) = (f \circ (ab)^{-1})(x) = \rho(ab)f(x).$$

Proposition 3.4.2

The action of $\mathbb{R}^n \rtimes O(n)$ on $L^2(\mathbb{R}^n)$ is unitary.

Proof:

Let $\alpha(x) = Ax + T$ with $A \in O(n)$, and substitute

$$y = \alpha^{-1}(x), \quad x = \alpha(y) = Ay + T.$$

Since $|\det A| = 1$, we have

$$d^n x = d^n y,$$

and therefore

$$\langle \rho(\alpha)f \mid \rho(\alpha)g \rangle = \int \overline{f(\alpha^{-1}(x))} g(\alpha^{-1}(x)) d^n x = \int \overline{f(y)} g(y) d^n y = \langle f \mid g \rangle.$$

Definition 3.4.3: Hamiltonian

A **Hamiltonian** is a self-adjoint, usually unbounded, operator

$$H : \mathcal{H} \rightarrow \mathcal{H}.$$

Definition 3.4.4: Spatial Symmetries of the Hamiltonian

The **group of spatial symmetries** of the Hamiltonian H is the subgroup

$$\text{Sym}(H) = \{a \in \mathbb{R}^n \rtimes O(n) \mid [\rho(a), H] = 0\}.$$

Remark.

Since $\rho(a)H = H\rho(a)$ for every $a \in \text{Sym}(H)$, the Hamiltonian is an intertwiner of the representation of $\text{Sym}(H)$ on $\mathcal{H}$ with itself.

Assume for the moment that the representation of $\text{Sym}(H)$ on $\mathcal{H}$ decomposes as

$$\mathcal{H} = \bigoplus_i N_i V_i,$$

where the V_i are pairwise non-isomorphic irreducible representations and $N_i V_i$ is shorthand for the isotypic component $N_i \otimes V_i$, with N_i the multiplicity space. From Schur's lemma,

$$\langle \psi_i \mid H \varphi_j \rangle = 0 \quad \text{for} \quad \psi_i \in N_i V_i, \quad \varphi_j \in N_j V_j, \quad i \neq j.$$

Remark.

Why does this work here? By the Peter-Weyl theorem, unitary representations of compact groups decompose into irreducible representations. Thus for compact symmetry groups, such as $O(n)$ or $SO(3)$, the direct-sum decomposition above is justified.

If the symmetry group contains a non-compact part, such as translations $\mathbb{R}^n$, the correct analogue is usually a direct integral rather than a direct sum. This is exactly what happens in Fourier theory and in the Bloch decomposition for lattice translations below.

It follows that H is block diagonal with respect to the above decomposition. Moreover, Schur's

lemma forces H to act as the identity on the irreducible factor V_i , hence

$$H = \bigoplus_i H_i \otimes I_{V_i}.$$

Example.

- **Free particle.** The Hamiltonian

$$H = -\frac{\hbar^2}{2m} \nabla^2$$

is invariant under translations and orthogonal transformations, so

$$\text{Sym}(H) = \mathbb{R}^n \rtimes O(n).$$

- **Particle in a potential.** For

$$H = -\frac{\hbar^2}{2m} \nabla^2 + V(x),$$

the multiplication operator V acts on wave functions by

$$(Vf)(x) = V(x)f(x),$$

so that, for $a = \alpha \in \mathbb{R}^n \rtimes O(n)$,

$$\rho(a)(Vf)(x) = V(\alpha^{-1}(x)) f(\alpha^{-1}(x)),$$

while

$$V(\rho(a)f)(x) = V(x) f(\alpha^{-1}(x)).$$

Consequently,

$$[\rho(a), V]f(x) = (V(\alpha^{-1}(x)) - V(x)) f(\alpha^{-1}(x)).$$

Hence a is a symmetry if

$$V(\alpha^{-1}(x)) = V(x) \quad \forall x \in \mathbb{R}^n,$$

and so

$$\text{Sym}(H) = \{a \in \mathbb{R}^n \rtimes O(n) \mid V(ax) = V(x) \ \forall x \in \mathbb{R}^n\}.$$

- **Hydrogen atom.** The Coulomb potential

$$V(x) \sim \frac{1}{\|x\|}$$

is invariant under rotations, since

$$\|Rx\| = \|x\| \quad \forall R \in O(3).$$

The Hilbert space decomposes as

$$\mathcal{H} \cong \bigoplus_{\ell=0}^{\infty} L^2(\mathbb{R}_{>0}, r^2 dr) \otimes \mathcal{H}_{\ell},$$

where $\mathcal{H}_{\ell}$ is the space of spherical harmonics of degree ℓ ,

$$\mathcal{H}_{\ell} = \text{span}\{Y_{\ell}^m : m = -\ell, \dots, \ell\}.$$

This is an irreducible representation of $SO(3)$ of dimension

$$\dim \mathcal{H}_{\ell} = 2\ell + 1.$$

From Schur's lemma,

$$\langle \psi_{\ell, m} | H | \varphi_{\ell', m'} \rangle = 0 \quad \text{if } \ell' \neq \ell.$$

Thus angular momentum labels the irreducible representations.

- **Lattices and Bloch decomposition.** Let $\Delta \subseteq \mathbb{R}^n$ be a lattice, and consider a periodic potential

$$V(x + T) = V(x) \quad \forall T \in \Delta.$$

Then H has at least the symmetry group Δ . Assuming $\Delta \cong \mathbb{Z}^n$, choose a basis $\{a_1, \dots, a_n\}$ of lattice vectors, so that every lattice vector can be written as

$$T = m_1 a_1 + \dots + m_n a_n, \quad m_i \in \mathbb{Z}.$$

Since $\Delta \cong \mathbb{Z}^n$ is abelian, all of its irreducible representations are one-dimensional, i.e. group homomorphisms

$$\mathbb{Z}^n \rightarrow U(1).$$

The group of such homomorphisms is

$$\text{Hom}(\mathbb{Z}^n, U(1)) \cong U(1)^n.$$

A convenient way to write these representations is by choosing plane waves: for $k \in \mathbb{R}^n$, consider

$$e^{2\pi i k \cdot x}.$$

Then

$$e^{2\pi i k \cdot (x-T)} = e^{2\pi i k \cdot x} e^{-2\pi i k \cdot T},$$

so that the character of Δ is

$$\chi_k(T) = e^{-2\pi i k \cdot T}.$$

In particular,

$$\chi_k(a_j) = e^{-2\pi i k \cdot a_j},$$

and the corresponding element of $U(1)^n$ is

$$(e^{-2\pi i k \cdot a_1}, \dots, e^{-2\pi i k \cdot a_n}).$$

Different values of k can yield the same character: k and $k + G$ give the same character

if and only if

$$e^{-2\pi i G \cdot T} = 1 \quad \forall T \in \Delta,$$

that is,

$$G \cdot T \in \mathbb{Z} \quad \forall T \in \Delta.$$

Definition 3.4.5: Reciprocal Lattice and Brillouin Zone

The **reciprocal lattice** is

$$\Delta^* = \{G \in \mathbb{R}^n \mid G \cdot T \in \mathbb{Z} \ \forall T \in \Delta\}.$$

Equivalently, if $\{a_1, \dots, a_n\}$ is a basis of Δ , then the dual basis $\{a_1^*, \dots, a_n^*\}$ is defined by

$$a_i^* \cdot a_j = \delta_{ij},$$

and the possible characters are labeled by

$$k \in \mathbb{R}^n / \Delta^*.$$

This quotient, or a fundamental domain for it, is called the **Brillouin zone**.

Since the Hamiltonian commutes with lattice translations, the Hilbert space decomposes into sectors labeled by k :

$$\mathcal{H} = \int_{\text{BZ}}^{\oplus} \mathcal{H}_k dk,$$

where $\mathcal{H}_k$ is the subspace transforming under translations as χ_k .

3.4.2 Selection Rules

In quantum mechanics, one considers both states

$$|\psi\rangle \in \mathcal{H}$$

and operators on $\mathcal{H}$. Observables are represented by Hermitian operators O , so that

$$\langle O\psi \mid \varphi \rangle = \langle \psi \mid O\varphi \rangle.$$

In particular, expectation values

$$\langle \psi \mid O\psi \rangle$$

are real. If O is positive, then

$$\langle \psi \mid O\psi \rangle \geq 0.$$

Suppose G is a symmetry group of the system. Then $\mathcal{H}$ carries a unitary representation

$$\rho : G \rightarrow U(\mathcal{H}).$$

The induced action on operators is by conjugation:

$$O \longmapsto \rho(g)O\rho(g^{-1}).$$

For time evolution, the Heisenberg-picture operator is

$$O(t) = e^{-iHt}O(0)e^{iHt}, \quad t \in \mathbb{R}.$$

Let $|\psi\rangle \in \mathcal{H}$. Then the transformed vector $O|\psi\rangle$ transforms as

$$O|\psi\rangle \mapsto \rho(g)O\rho(g^{-1})\rho(g)|\psi\rangle = \rho(g)(O|\psi\rangle).$$

Abstractly, this means that the evaluation map

$$\text{End}(\mathcal{H}) \otimes \mathcal{H} \longrightarrow \mathcal{H}, \quad O \otimes |\psi\rangle \mapsto O|\psi\rangle,$$

is G -equivariant.

Proposition 3.4.6

Suppose O transforms in an irreducible representation $W \subseteq \text{End}(\mathcal{H})$, and suppose $|\psi_i\rangle$ and $|\psi_j\rangle$ transform in irreducible representations $V_i, V_j \subseteq \mathcal{H}$. If V_j does not occur in the decomposition

$$W \otimes V_i \cong \bigoplus_k N_k,$$

then

$$\langle \psi_j | O\psi_i \rangle = 0.$$

Proof:

Since O transforms in W and $|\psi_i\rangle$ transforms in V_i , the vector $O|\psi_i\rangle$ lies in the representation generated by $W \otimes V_i$. If V_j is not one of the irreducible summands of $W \otimes V_i$, then $O|\psi_i\rangle$ has no component in the V_j -isotypic part of $\mathcal{H}$. Orthogonality of distinct irreducible components gives

$$\langle \psi_j | O\psi_i \rangle = 0.$$

Example.

Consider the ammonia molecule NH_3 . Its spatial symmetry group is S_3 , permuting the three hydrogen atoms. The irreducible representations of S_3 are

$$\text{Triv}, \quad \text{Alt}, \quad \text{Std}.$$

Geometrically, the z -axis transforms trivially, while the x - and y -directions transform in the standard representation:

$$\hat{z} \in \text{Triv}, \quad \hat{x}, \hat{y} \in \text{Std}.$$

We apply the selection rule to the coordinate operator $\hat{y}$, so $W = \text{Std}$. Let $|\psi_i\rangle \in V_i$. Then $\hat{y}|\psi_i\rangle$ transforms in the tensor product

$$\text{Std} \otimes V_i.$$

Hence:

- If $V_i = \text{Triv}$, then

$$\text{Std} \otimes \text{Triv} \cong \text{Std}.$$

Therefore

$$\langle \psi_j | \hat{y} | \psi_i \rangle = 0$$

unless $|\psi_j\rangle$ transforms in Std.

- If $V_i = \text{Alt}$, then

$$\text{Std} \otimes \text{Alt} \cong \text{Std}.$$

Again,

$$\langle \psi_j | \hat{y} | \psi_i \rangle = 0$$

unless $|\psi_j\rangle$ transforms in Std.

- If $V_i = \text{Std}$, then

$$\text{Std} \otimes \text{Std} \cong \text{Triv} \oplus \text{Alt} \oplus \text{Std}.$$

Thus the matrix element

$$\langle \psi_j | \hat{y} | \psi_i \rangle$$

is not ruled out by symmetry.

Example.

Multi-particle states. Suppose $\mathcal{H}$ is the one-particle Hilbert space. For a system of N particles, the natural Hilbert space is

$$\mathcal{H}^{\otimes N}.$$

The symmetric group S_N acts on $\mathcal{H}^{\otimes N}$ by permuting the tensor factors.

In a quantum system of indistinguishable elementary particles, exchanging two particles can only change the state by a phase:

$$e^{i\theta} |\psi\rangle = P_{(12\dots N)} |\psi\rangle.$$

Hence the multi-particle state lies in a one-dimensional representation of S_N . There are only two such representations:

$$\text{Triv} \quad \text{and} \quad \text{Alt}.$$

The trivial representation gives bosons, while the alternating representation gives fermions. Thus

$$\mathcal{H}_{\text{bos}} = \text{Sym}^N(\mathcal{H}) \subseteq \mathcal{H}^{\otimes N}, \quad \mathcal{H}_{\text{ferm}} = \bigwedge^N \mathcal{H} \subseteq \mathcal{H}^{\otimes N}.$$

If $\dim \mathcal{H} = k$, then

$$\dim \text{Sym}^N(\mathcal{H}) = \binom{k+N-1}{N}, \quad \dim \bigwedge^N \mathcal{H} = \binom{k}{N}.$$

For a spin- $\frac{1}{2}$ system,

$$\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2,$$

where $L^2(\mathbb{R}^3)$ is the orbital part and $\mathbb{C}^2 = \text{span}\{|\uparrow\rangle, |\downarrow\rangle\}$ is the spin part. Since spin- $\frac{1}{2}$

particles are fermions, the physical N -particle Hilbert space is

$$\bigwedge^N (L^2(\mathbb{R}^3) \otimes \mathbb{C}^2).$$

For $N = 2$, the spin states decompose as

$$(\mathbb{C}^2)^{\otimes 2} = \text{Sym}^2 \mathbb{C}^2 \oplus \bigwedge^2 \mathbb{C}^2.$$

The symmetric spin states are

$$|\uparrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle, \quad \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle),$$

and the antisymmetric spin state is

$$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

Therefore

$$\mathcal{H} = L^2(\mathbb{R}^3 \times \mathbb{R}^3) \otimes (\text{Sym}^2 \mathbb{C}^2 \oplus \bigwedge^2 \mathbb{C}^2),$$

but the physical subspace is still the fermionic subspace.

For $N = 3$,

$$(\mathbb{C}^2)^{\otimes 3} \cong \mathbb{C}^8.$$

The fully symmetric spin states include

$$|\uparrow\uparrow\uparrow\rangle, \quad |\downarrow\downarrow\downarrow\rangle, \quad \frac{1}{\sqrt{3}}(|\uparrow\uparrow\downarrow\rangle + |\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle),$$

and

$$\text{Sym}^3 \mathbb{C}^2 \cong \text{Triv}^{\oplus 4}.$$

On the other hand,

$$\bigwedge^3 \mathbb{C}^2 = 0$$

by the Pauli exclusion principle.

To find the missing two-dimensional pieces, one uses the irreducible representations of S_3 . For example, for the Young diagram

$$T = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array},$$

the row group and column group are

$$R_T = \{e, (12)\}, \quad C_T = \{e, (13)\}.$$

Hence the Young symmetrizer is

$$\gamma_T = a_T b_T = \left(\sum_{\sigma \in R_T} \sigma \right) \left(\sum_{\tau \in C_T} \text{sgn}(\tau) \tau \right) = e + (12) - (13) - (132).$$

The Specht module

$$S_T := \mathbb{C}[S_3]\gamma_T$$

is the standard representation. Applying Gram-Schmidt to γ_T and $(13)\gamma_T$ gives a basis

$$\gamma_T = \frac{1}{2}(e + (12) - (13) - (132)),$$

$$\gamma_T^\perp = \frac{1}{2\sqrt{3}}(-2e + (12) - (13) - (132)).$$

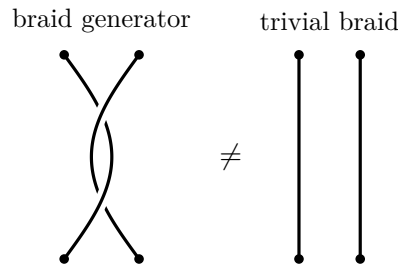
Acting with these elements on $|\uparrow\uparrow\downarrow\rangle$ gives states in the standard representation:

$$\gamma_T|\uparrow\uparrow\downarrow\rangle = \frac{1}{\sqrt{6}}(2|\uparrow\uparrow\downarrow\rangle - |\downarrow\uparrow\uparrow\rangle - |\uparrow\downarrow\uparrow\rangle),$$

$$\gamma_T^\perp|\uparrow\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}}(|\downarrow\uparrow\uparrow\rangle - |\uparrow\downarrow\uparrow\rangle).$$

Remark.

Fermions and bosons only exhaust the possibilities in spacetime dimension at least 3. In two spatial dimensions, exchanges are represented by the braid group B_N , not just by S_N . Thus particles in two dimensions can be anyons.



Chapter 4

Lie groups

"Groups with a notion of continuity."

Example.

- $(\mathbb{R}, +)$ has a notion of closeness $\lim_{n \rightarrow \infty} a_n = b$ and a notion of continuity as we can describe paths $\gamma(-\epsilon, \epsilon) \rightarrow \mathbb{R}$
- $(\mathbb{R}^n, +)$, $U(N)$, $SU(N)$, $SO(N)$, $GL(N, \mathbb{R})$, $GL(N, \mathbb{C})$ are all groups with a notion of closeness and continuity

Consider the 1 dimensional representations of $\mathbb{R}$, i.e. the homomorphisms

$$\rho : (\mathbb{R}, +) \rightarrow (\mathbb{C}^\times, \cdot)$$

Given $\rho(1) = c$, we can immediately conclude the following:

- $\rho(n) = \rho(1 + 1 + \dots + 1) = \rho(1)^n = c^n$
- $c = \rho(\frac{1}{n} \cot n) = \rho(\frac{1}{n})^n \Rightarrow \rho(\frac{1}{n}) = c^{\frac{1}{n}}$
- $\rho(\frac{m}{n}) = c^{\frac{m}{n}}$

Since $\mathbb{R}$ is a vector space over $\mathbb{Q}$, it has a $\mathbb{Q}$ -basis $\{1, a_i\}_{i \in I}$ with uncountable index set I . Then for a representation ρ , we have $\rho(a_i) \in \mathbb{C}^\times$ and $\rho(\sum_{i \in I} \lambda_i a_i) = \prod_{i \in I} \rho(a_i)^{\lambda_i}$ for $\lambda_i \in \mathbb{Q}$. This means that the representation is determined by the values $\{\rho(a_i)\}_{i \in I}$, which can be chosen arbitrarily. Hence, there are uncountably many 1-dimensional representations of $\mathbb{R}$. Thus representations without continuity conditions are hopeless.

Definition 4.0.1: Matrix Lie groups

A **matrix Lie group** is a closed subgroup of $GL(n, \mathbb{R})$. Closedness is with respect to the usual topology on $\mathbb{R}^{n^2}$.

Remark.

Complex matrix groups e.g. $SU(n)$ can be viewed as a subgroup of $GL(2n, \mathbb{R})$.

Example.

The real numbers with addition $(\mathbb{R}, +) \cong \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} : x \in \mathbb{R} \right\}$ is a matrix Lie group.

4.1 A few basics from differential geometry

Definition 4.1.1: topological space

A **topological space** X is characterized by a collection of subsets called the opensets, $\mathcal{O}(X) \subseteq \mathcal{P}(X)$ such that:

- $\emptyset, X \in \mathcal{O}(X)$
- $\bigcup_{i \in I} U_i \in \mathcal{O}(X)$ for U_i a family of opensets
- $U_1 \cap U_2 \in \mathcal{O}(X)$ for $U_1, U_2 \in \mathcal{O}(X)$

Example.

Let (X, d) be a metric space. Define $U \subseteq X$ to be open if

$$\forall x \in U \quad \exists \epsilon > 0 \text{ such that } B(x, \epsilon) := \{y \in X : d(y, x) < \epsilon\} \subseteq U$$

Definition 4.1.2: continuous map

A map $f : X \rightarrow Y$ between topological spaces is **continuous** if

$$f^{-1}(U) \in \mathcal{O}(X) \quad \forall U \in \mathcal{O}(Y).$$

If f is a bijection whose inverse is also continuous, it is called a **homeomorphism**.

Definition 4.1.3: chart

A **chart** on a topological space X is a pair (U_α, ψ_α) consisting of an open set $U_\alpha \subseteq X$ and a homeomorphism

$$\psi_\alpha : U_\alpha \rightarrow V_\alpha \subseteq \mathbb{R}^n.$$

For two charts (U_α, ψ_α) and (U_β, ψ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the **transition map**

$$\psi_{\alpha\beta} : \tilde{V}_\alpha \rightarrow \tilde{V}_\beta$$

is obtained by restricting ψ_α, ψ_β to the images $\tilde{V}_\alpha, \tilde{V}_\beta$ of the intersection $U_\alpha \cap U_\beta$. Since this is a map between subsets of $\mathbb{R}^n$, the notion of smoothness is well defined.

Definition 4.1.4: atlas

An **atlas** on X is a collection of charts $\{(U_i, \psi_i)\}_{i \in I}$ whose domains form an open cover of X :

$$\bigcup_{i \in I} U_i = X.$$

An atlas is **maximal** if it is not contained in a larger atlas.

Definition 4.1.5: Hausdorff space

A topological space X is called **Hausdorff** if

$$\forall x, y \in X \text{ with } x \neq y \quad \exists U_x, U_y \in \mathcal{O}(X) \text{ such that } x \in U_x, y \in U_y, U_x \cap U_y = \emptyset.$$

Definition 4.1.6: manifold

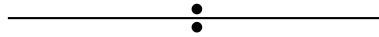
An n -dimensional (smooth) manifold is a topological Hausdorff space equipped with a maximal atlas.

Remark.

Why Hausdorff? We want unique limits. Consider

$$\mathbb{R} \sqcup \mathbb{R} / (x_1 \sim x_2 \text{ for } x_1 \neq 0)$$

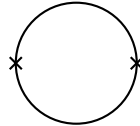
→ “ $\mathbb{R}$ with two zeros”.



This has a maximal atlas but is *not* Hausdorff. Limits are not uniquely defined. Thus we require Hausdorffness to ensure that limits are well-defined.

Example.

- $\mathbb{R}^n$ is a manifold; $\text{id} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an atlas. (This manifold structure is unique, except $\mathbb{R}^4$ which has ∞ many.)
- $S^1 \subset \mathbb{R}^2$ defined by $x^2 + y^2 = 1$ is covered by the four charts



$$\psi_{\pm}^x : \{(x, \pm\sqrt{1-x^2}) \mid x \neq \pm 1\} \longrightarrow (-1, 1), \quad (x, y) \mapsto x$$

$$\psi_{\pm}^y : \{(\pm\sqrt{1-y^2}, y) \mid y \neq \pm 1\} \longrightarrow (-1, 1), \quad (x, y) \mapsto y$$

$\Rightarrow S^1$ is a manifold.

Definition 4.1.7: smooth map

Let M, N be manifolds. A continuous map $f : M \rightarrow N$ is **smooth** if for all $m \in M$ there exists a chart U_m containing m and a chart $U_{f(m)}$ containing $f(U_m)$ such that

$$\psi_{f(m)} \circ f \circ \psi_m^{-1}$$

is smooth. A **diffeomorphism** is a smooth map with smooth inverse.

Definition 4.1.8: derivation

For M a manifold, we write $C^\infty(M)$ for the vector space of smooth functions $f : M \rightarrow \mathbb{R}$. For $x \in M$, a **derivation** at x is a linear map

$$D_x : C^\infty(M) \longrightarrow \mathbb{R}$$

satisfying the Leibniz rule

$$D_x(f \cdot g) = D_x(f) \cdot g(x) + f(x) \cdot D_x(g).$$

Given local coordinates $(x_1, \dots, x_n)$ on a chart U_α containing x , one defines

$$\left. \frac{\partial}{\partial x_i} \right|_x : C^\infty(M) \longrightarrow \mathbb{R}, \quad f \longmapsto \left. \frac{\partial}{\partial x_i} \right|_{\psi_\alpha(x)} (f \circ \psi_\alpha^{-1}).$$

The derivations $\left. \frac{\partial}{\partial x_i} \right|_x$ form a basis of the vector space of derivations at x .

Definition 4.1.9: tangent space

The **tangent space** to M at $x \in M$ is the vector space

$$T_x M := \{\text{derivations at } x\}.$$

Example.

- $T_x \mathbb{R}^n = \text{span} \left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right\}$
- $T_v V = V$ for V a vector space.

Remark.

Tangent vectors as equivalence classes of curves. Consider a smooth curve

$$\gamma(t) : (-\varepsilon, \varepsilon) \longrightarrow M, \quad \gamma(0) = x.$$

Its differential at 0 is a linear map

$$d\gamma|_0 : T_0(-\varepsilon, \varepsilon) \longrightarrow T_x M, \quad \frac{\partial}{\partial t} \longmapsto d\gamma|_0 \left[\frac{\partial}{\partial t} \right].$$

For $F \in C^\infty(M)$ this acts as

$$d\gamma|_0 \left[\frac{\partial}{\partial t} \right] (F) = \left. \frac{\partial}{\partial t} \right|_{t=0} F(\gamma(t)).$$

For $M = V$ a vector space with $T_x V = V$, writing $\gamma(t) = \gamma^1(t)x^1 + \dots + \gamma^n(t)x^n$ gives

$$\frac{\partial}{\partial t} F(\gamma(t)) = \frac{\partial F}{\partial x^1} \frac{\partial \gamma^1}{\partial t} + \dots + \frac{\partial F}{\partial x^n} \frac{\partial \gamma^n}{\partial t},$$

so that

$$d\gamma|_0 \left[\frac{\partial}{\partial t} \right] = \dot{\gamma}(0) \in V = T_x V.$$

Hence a tangent vector at x can equivalently be viewed as the velocity $\dot{\gamma}(0)$ of a curve through x .

Definition 4.1.10: differential

Let $f : M \rightarrow N$ be a smooth map. It induces a linear map

$$df|_x : T_x M \longrightarrow T_{f(x)} N, \quad D_x \longmapsto \left(h \mapsto D_x(h \circ f) \right) \quad \forall h \in C^\infty(N),$$

called the **differential**, or **push-forward**, of f . In local coordinates x^i on M and y^j on N ,

$$\frac{\partial}{\partial x^i} \longmapsto \sum_j \frac{\partial f^j}{\partial x^i} \frac{\partial}{\partial y^j}.$$

4.2 Lie algebras

Definition 4.2.1: Lie group

A **Lie group** is a smooth manifold G equipped with a group structure such that

$$\bullet : G \times G \longrightarrow G \quad \text{and} \quad (\cdot)^{-1} : G \longrightarrow G$$

are smooth maps.

Remark.

The Cartesian product of manifolds is (in most cases) also a manifold.

G acts on itself by conjugation

$$\text{Ad}(g) : G \longrightarrow G, \quad h \longmapsto ghg^{-1},$$

which is a smooth map since Lie groups act smoothly by definition. Taking the differential at the identity gives a linear map

$$d\text{Ad}(g)|_e : T_e G \longrightarrow T_e G,$$

and hence a group homomorphism

$$\text{ad} : G \longrightarrow \text{Aut}(T_e G).$$

Definition 4.2.2: smooth representation

Let G be a Lie group. A **(smooth) representation** of G is a smooth group homomorphism

$$G \longrightarrow \text{Aut}(V)$$

for a vector space V .

Proposition 4.2.3

Let G be a Lie group. The map

$$\text{ad} : G \longrightarrow \text{Aut}(T_e G), \quad g \longmapsto d\text{Ad}(g)|_e,$$

is a representation of G , called the **adjoint action**.

Proof:

Homomorphism. We need $\text{ad}(g) \circ \text{ad}(h) = \text{ad}(gh)$, i.e.

$$d\text{Ad}(g)|_e \circ d\text{Ad}(h)|_e = d\text{Ad}(gh)|_e.$$

This follows from $\text{Ad}_g \circ \text{Ad}_h = \text{Ad}_{gh}$ together with the chain rule: for smooth maps $f : M \rightarrow N$ and $f' : N \rightarrow O$,

$$df'|_{f(m)} \circ df|_m = d(f' \circ f)|_m.$$

Indeed, for a derivation $D_m \in T_m M$ and $h \in C^\infty(O)$,

$$\begin{aligned} (df'|_{f(m)} \circ df|_m)(D_m)[h] &= df'|_{f(m)}(D_m(h \circ f' \circ \cdot)) \\ &= D_m(h \circ f' \circ f) = d(f' \circ f)|_m(D_m)[h]. \end{aligned}$$

Identity. $\text{Ad}(e) = \text{id}_G$, so $\text{ad}(e) = d(\text{id}_G)|_e = \text{id}_{T_e G}$.

Smoothness. Follows from the smoothness of the group operations. ■

Proposition 4.2.4

Let $f : G \rightarrow H$ be a Lie group homomorphism. Then $df|_e : T_e G \rightarrow T_e H$ satisfies

$$df|_e \circ \text{ad}_G(g) = \text{ad}_H(f(g)) \circ df|_e \quad \forall g \in G.$$

Proof:

We first show that $f \circ \text{Ad}_G(g) = \text{Ad}_H(f(g)) \circ f$. For all $g, h \in G$,

$$f(\text{Ad}_G(g)[h]) = f(ghg^{-1}) = f(g)f(h)f(g)^{-1} = \text{Ad}_H(f(g))[f(h)].$$

The statement then follows by differentiating both sides at the identity using $d(\cdot)|_e$. ■

Example.

Let V be a finite-dimensional vector space. The group $\text{Aut}(V)$ is an open subset of $\text{End}(V) \cong \mathbb{R}^{2 \dim V}$, so its tangent space at the identity is

$$T_{\text{id}} \text{Aut}(V) \cong \text{End}(V).$$

The adjoint action of $\text{Aut}(V)$ on $\text{End}(V)$ is

$$\text{ad}(U) : A \mapsto UAU^{-1}.$$

Definition 4.2.5: Lie bracket

Let G be a Lie group. The differential of ad at the identity gives a map

$$d\text{ad}|_e : T_e G \longrightarrow T_{\text{id}} \text{Aut}(T_e G) = \text{End}(T_e G).$$

The **Lie bracket** on $T_e G$ is defined by

$$[X, Y] := d\text{ad}|_e(X)[Y] \in T_e G \quad \forall X, Y \in T_e G.$$

How do we compute $[X, Y]$?

Example.

Consider $G = \text{Aut}(V)$, so $T_e G = \text{End}(V)$. For $X \in \text{End}(V)$, the curve

$$\gamma_X(t) = \mathbb{I} + tX$$

passes through e at $t = 0$ with tangent vector X , since $X[v] = \frac{\partial}{\partial t}|_{t=0} \gamma_X(t)[v]$.

By definition of the Lie bracket,

$$[X, Y]F = d\text{ad}|_e(X)[Y]F = \frac{\partial}{\partial t_1} \Big|_{t_1=0} \text{ad}(\gamma_X(t_1))[Y]F.$$

Since $\text{ad}(U)$ acts by conjugation, $\text{ad}(\gamma_X(t_1))[Y]F = Y[F(\gamma_X(t_1) \cdot \gamma_X(t_1)^{-1})]$. Representing Y by a curve γ_Y ,

$$\begin{aligned} [X, Y]F &= \frac{\partial}{\partial t_1} \Big|_{t_1=0} Y[F(\gamma_X(t_1) (\cdot) \gamma_X(t_1)^{-1})] \\ &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} F(\gamma_X(t_1) \gamma_Y(t_2) \gamma_X(t_1)^{-1}). \end{aligned} \quad (4.1)$$

For $G = \text{GL}(n)$ with $\gamma_X(t) = \mathbb{I} + t_1 X$ and $\gamma_Y(t) = \mathbb{I} + t_2 Y$,

$$\begin{aligned} [X, Y] &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} (\mathbb{I} + t_1 X)(\mathbb{I} + t_2 Y)(\mathbb{I} + t_1 X)^{-1} \\ &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} (\mathbb{I} + t_1 X)(\mathbb{I} + t_2 Y)(\mathbb{I} - t_1 X + \mathcal{O}(t_1^2)) \\ &= XY - YX. \end{aligned}$$

We have shown that any Lie group G gives rise to a tangent space $T_e G$ equipped with a bilinear

bracket

$$[-, -] : T_e G \otimes T_e G \longrightarrow T_e G,$$

and that for matrix Lie groups this bracket is the commutator $[X, Y] = XY - YX$.

Proposition 4.2.6

Let $f : G \rightarrow H$ be a Lie group homomorphism. Then

$$[df|_e X, df|_e Y]_{T_e H} = df|_e [X, Y]_{T_e G}.$$

Proof:

Using formula (4.1),

$$\begin{aligned} [df|_e X, df|_e Y]_{T_e H} F &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} F \left(f(\gamma_X(t_1)) f(\gamma_Y(t_2)) f(\gamma_X(t_1))^{-1} \right) \\ &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} F \left(f \left(\gamma_X(t_1) \gamma_Y(t_2) \gamma_X(t_1)^{-1} \right) \right) \\ &= df|_e [X, Y]_{T_e G} F, \end{aligned}$$

where in the second step we used that f is a group homomorphism, and in the third step we applied formula (4.1) for G . ■

Remark.

For matrix Lie groups $G \leq \text{GL}(n)$, the inclusion $G \hookrightarrow \text{GL}(n)$ is a Lie group homomorphism, so $T_e G \subseteq \text{End}(\mathbb{R}^n)$ and the bracket on $T_e G$ is the restriction of the commutator:

$$[X, Y]_{T_e G} = XY - YX.$$

Theorem 4.2.7: Properties of the Lie bracket

Let G be a Lie group. For all $X, Y, Z \in T_e G$:

1. $[X, Y] = -[Y, X]$ (antisymmetric)
2. $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ (Jacobi identity)

Proof:

Antisymmetry. We show $[X, X] = 0$; antisymmetry then follows from

$$0 = [X + Y, X + Y] = [X, Y] + [Y, X].$$

Assume $\gamma_X(t) \cdot \gamma_X(t') = \gamma_X(t + t')$ for t, t' near 0. Then in local coordinates

$$\frac{\partial}{\partial t} \gamma(t) = \lim_{\varepsilon \rightarrow 0} \frac{\gamma(t + \varepsilon) - \gamma(t)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \gamma(t) \frac{\gamma(\varepsilon) - \gamma(0)}{\varepsilon} = \gamma(t) \cdot X.$$

This is a system of ODEs and hence has a unique solution near $t = 0$, so γ_X exists. Using

the Proposition above,

$$[X, X] F = \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{\substack{t_1=0 \\ t_2=0}} F(\gamma_X(t_1) \gamma_X(t_2) \gamma_X(-t_1)) = \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{\substack{t_1=0 \\ t_2=0}} F(\gamma_X(t_2)) = 0,$$

where we used $\gamma_X(t_1) \gamma_X(t_2) \gamma_X(-t_1) = \gamma_X(t_2)$.

Jacobi identity. Consider

$$H : T_e G \rightarrow \text{End}(T_e G), \quad X \mapsto [X, -],$$

which is the differential of $G \rightarrow \text{Aut}(T_e G)$. By the previous proposition,

$$H([X, Y]) := [[X, Y], -] = H(X) \circ H(Y) - H(Y) \circ H(X).$$

Evaluating at $Z \in T_e G$:

$$\begin{aligned} [[X, Y], Z] &= [X, [Y, Z]] - [Y, [X, Z]] \\ &= -[Z, [X, Y]] = [X, [Y, Z]] + [Y, [Z, X]], \end{aligned}$$

hence $0 = [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]]$. ■

Definition 4.2.8: Lie algebra

A **Lie algebra** is a vector space $\mathfrak{g}$ equipped with a linear map

$$[-, -] : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$$

such that for all $X, Y, Z \in \mathfrak{g}$:

- $[X, Y] = -[Y, X]$
- $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ (Jacobi identity)

Definition 4.2.9: Lie algebra homomorphism

A **Lie algebra homomorphism** is a map $f : \mathfrak{g} \rightarrow \mathfrak{g}'$ such that

$$f([X, Y]) = [f(X), f(Y)].$$

Remark.

For any Lie group, $T_e G =: \mathfrak{g}$ is a Lie algebra, and every Lie group homomorphism $G \rightarrow H$ induces a Lie algebra homomorphism $\mathfrak{g} \rightarrow \mathfrak{h}$. In particular, a representation $\rho : G \rightarrow \text{Aut}(V)$ induces a Lie algebra homomorphism $\mathfrak{g} \rightarrow \text{End}(V)$.

Definition 4.2.10: Representation of a Lie algebra

Let $\mathfrak{g}$ be a Lie algebra. A **representation** of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\mathfrak{g} \rightarrow \text{End}(V)$$

for a vector space V .

Remark.

Most definitions for finite groups, such as irreducible representations, carry over unchanged to Lie groups. We will not repeat them here.

Example.

The Lie group $SU(2)$ and its Lie algebra $\mathfrak{su}(2)$.

$$SU(2) := \{U \in GL(2, \mathbb{C}) \mid U^\dagger U = I, \det(U) = 1\}$$

This is not a complex Lie group (real dimension 3), even though the elements are complex.

For $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $U^\dagger = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}$,

$$U^\dagger U = \begin{pmatrix} \bar{a}a + \bar{c}c & \bar{a}b + \bar{c}d \\ \bar{b}a + \bar{d}c & \bar{b}b + \bar{d}d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad ad - bc = 1.$$

All solutions are of the form

$$\begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (**)$$

hence $\dim SU(2) = 3$.

For every $U \in SU(2)$, there exists $V \in U(2)$ such that

$$VUV^{-1} = \begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix}.$$

Setting $V' = (\det V)^{-1/2}V$ gives $V' \in SU(2)$, so the conjugacy classes are labelled by $\alpha \in U(1)/\mathbb{Z}_2 \cong \mathbb{R}/\pi\mathbb{R}$.

Writing $\alpha = a + ib$ and $\beta = c + id$ with $a, b, c, d \in \mathbb{R}$, condition $(**)$ gives

$$a^2 + b^2 + c^2 + d^2 = 1,$$

so $SU(2) \cong S^3$ as a manifold. With this parameterisation,

$$U = \begin{pmatrix} a + ib & c + id \\ c - id & a - ib \end{pmatrix} = \underbrace{a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{i}} + \underbrace{b \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}}_{\mathbf{j}} + \underbrace{c \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\mathbf{k}} + \underbrace{d \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}}_{\mathbf{l}},$$

so $SU(2)$ is also the group of unit quaternions.

Connection to $SO(3)$. The submanifold $a = 0$ is diffeomorphic to S^2 and preserved under conjugation, giving a map

$$SU(2) = \text{Spin}(3) \longrightarrow SO(3) \quad (\text{not a bijection}),$$

with kernel $\{1, -1\}$, and $SO(3) = SU(2)/\mathbb{Z}_2$ is a double cover. Since $SU(2)$ and $SO(3)$ differ only by -1 , which is not near the identity, the Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are the same.

The Lie algebra $\mathfrak{su}(2)$. $\mathfrak{su}(2)$ consists of traceless skew-Hermitian matrices ($A = -A^\dagger$), with basis

$$X = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

The bracket satisfies $[X, Y] = 2Z$ and cyclic permutations.

Exponential map. For $A \in \text{Mat}(2, \mathbb{C})$, the matrix exponential is $\exp(A) = \sum_{n=0}^{\infty} \frac{1}{n!} A^n$. For $A \in \mathfrak{su}(2)$ (so $A^\dagger = -A$ and $\text{tr}(A) = 0$):

- $\exp(A)^\dagger \exp(A) = \exp(A^\dagger) \exp(A) = \exp\left(\underbrace{A^\dagger + A}_{=0}\right) = \mathbb{I},$
- $\det(\exp(A)) = \exp(\text{tr}(A)) = 1,$

so $\exp: \mathfrak{su}(2) \rightarrow SU(2)$ is a smooth map. Setting $\gamma_A(t) := \exp(tA)$ satisfies $\gamma_A(t+t') = \gamma_A(t)\gamma_A(t')$. Since $(U^\dagger A U)^n = U^\dagger A^n U$, we have $U^\dagger \exp(A) U = \exp(U^\dagger A U)$, so every conjugacy-class representative

$$\begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix} = \exp \begin{pmatrix} ix & 0 \\ 0 & -ix \end{pmatrix}, \quad \alpha = e^{ix},$$

is in the image, hence $\exp: \mathfrak{su}(2) \rightarrow SU(2)$ is surjective.

For a representation $\rho: \mathfrak{su}(2) \rightarrow \text{End}(V)$, one can try to define $\rho(\exp(A)) := \exp(\rho(A))$ for $g = \exp(A)$, but this is not well defined in general.

Remark.

The local solution γ_X extends to all of $\mathbb{R}$ via

$$\hat{\gamma}_X: \mathbb{R} \rightarrow G, \quad \hat{\gamma}_X(t) := \gamma_X\left(\frac{t}{n}\right)^n,$$

which is a Lie group homomorphism.

Definition 4.2.11: 1-parameter subgroup

A **1-parameter subgroup** of G is a Lie group homomorphism $\mathbb{R} \rightarrow G$. These are in bijection with $\mathfrak{g}$.

Definition 4.2.12: Exponential map

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The **exponential map** is

$$\exp : \mathfrak{g} \rightarrow G, \quad X \mapsto \gamma_X(1) \in G.$$

For matrix Lie groups this coincides with the matrix exponential $\exp(A) = \sum_{i=0}^{\infty} \frac{A^i}{i!}$.

Remark.

The differential $d\exp|_0 : \mathfrak{g} \rightarrow \mathfrak{g}$ is the identity. This implies that $\exp$ is a diffeomorphism when restricted to small neighbourhoods of $0 \in \mathfrak{g}$ and $e \in G$.

Remark.

There is a formula for products in G near e in terms of the Lie bracket: the Baker-Campbell-Hausdorff formula.

Proposition 4.2.13

Let $\psi : G \rightarrow H$ be a Lie group homomorphism. Then $\psi \circ \exp_G = \exp_H \circ d\psi|_e$, i.e., the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{d\psi|_e} & \mathfrak{h} \\ \exp_G \downarrow & & \downarrow \exp_H \\ G & \xrightarrow{\psi} & H \end{array}$$

Proof:

Consider $\gamma_X : \mathbb{R} \rightarrow G$. Then $\psi \circ \gamma_X$ is a group homomorphism $\mathbb{R} \rightarrow H$ (as a composition of group homomorphisms) with tangent vector $d\psi|_e[X]$ at $t = 0$. By the bijection between 1-parameter subgroups and $\mathfrak{h}$, it follows that $\psi \circ \gamma_X = \gamma_{d\psi|_e[X]}$, and

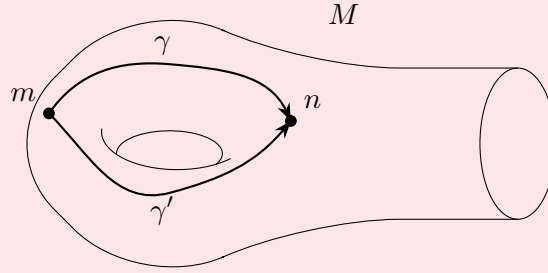
$$\psi(\exp(X)) = \psi(\gamma_X(1)) = \gamma_{d\psi|_e[X]}(1) = \exp(d\psi|_e[X]).$$

Definition 4.2.14: Path-connected and simply connected manifolds

A manifold M is **(path-) connected** if for each pair of points $m, n \in M$ there exists a continuous map $\gamma : [0, 1] \rightarrow M$ with $\gamma(0) = m$ and $\gamma(1) = n$.

M is called **simply connected** if for each pair of paths γ, γ' with $\gamma(0) = \gamma'(0) = m$ and $\gamma(1) = \gamma'(1) = n$, there exists a continuous map $H : [0, 1] \times [0, 1] \rightarrow M$ such that

- $H(0, t) = \gamma(t)$
- $H(1, t) = \gamma'(t)$
- $H(x, 0) = m$
- $H(x, 1) = n$

**Example.**

- $\mathbb{R}^n$ is simply connected.
- S^1 is connected but not simply connected.
- $SU(n)$ for $n \geq 2$ is simply connected.

Theorem 4.2.15: Lie's 1st Theorem

Let G be a connected Lie group. Two Lie group homomorphisms $\psi, \psi' : G \rightarrow H$ are equal if $d\psi|_e = d\psi'|_e$.

Proof:

By the local diffeomorphism property of $\exp$ and the identity $\psi \circ \exp_G = \exp_H \circ d\psi|_e$, the condition $d\psi|_e = d\psi'|_e$ implies $\psi = \psi'$ on a neighbourhood of e . It therefore suffices to show that every $g \in G$ is a product of elements from any neighbourhood U of e .

Let $\Gamma \subseteq G$ be the subgroup generated by U . Since $\Gamma = \bigcup_{h \in \Gamma} U \cdot h$, Γ is a union of open sets and hence open. Letting $\Gamma' = G \setminus \Gamma$, for any $h' \in \Gamma'$ the coset $U \cdot h' \subseteq \Gamma'$ (otherwise $u \cdot h' \in \Gamma$ would imply $h' = u^{-1}(u \cdot h') \in \Gamma$, a contradiction), so

$$\Gamma' = \bigcup_{h' \in \Gamma'} U \cdot h'$$

is likewise open. Thus $G = \Gamma \sqcup \Gamma'$ is a partition into two disjoint open sets.

If $\Gamma' \neq \emptyset$, pick $h \in \Gamma$ and $h' \in \Gamma'$. Since G is path-connected, there exists $\gamma : [0, 1] \rightarrow G$ with $\gamma(0) = h$ and $\gamma(1) = h'$. Then

$$[0, 1] = \gamma^{-1}(\Gamma) \cup \gamma^{-1}(\Gamma')$$

is a partition of $[0, 1]$ into two disjoint open non-empty sets, contradicting the connectedness of $[0, 1]$. Hence $\Gamma' = \emptyset$, i.e. $\Gamma = G$. ■

Theorem 4.2.16: Lie's 2nd Theorem

Let G be a simply connected Lie group. Then there is a one-to-one correspondence between Lie group homomorphisms $\psi : G \rightarrow H$ and Lie algebra homomorphisms $\Psi : \mathfrak{g} \rightarrow \mathfrak{h}$.

Theorem 4.2.17: Lie's 3rd Theorem

For each finite-dimensional real Lie algebra $\mathfrak{g}$ there exists a simply connected Lie group G with Lie algebra $\mathfrak{g}$.

Remark.

G in the above is unique: if G' is another simply connected Lie group with Lie algebra $\mathfrak{g}$, then $\text{id} : \mathfrak{g} \rightarrow \mathfrak{g}$ lifts to a diffeomorphism $G \xrightarrow{\sim} G'$.

Example. $(\mathbb{R}, +) \xrightarrow{\exp} U(1)$ is a Lie algebra isomorphism (both have Lie algebra $\mathbb{R}$), but the only Lie group homomorphism $U(1) \rightarrow \mathbb{R}$ is the constant map at 0, since $U(1) \cong S^1$ is not simply connected.

Corollary 4.2.18

For a simply connected Lie group G there is a bijection between representations of G and representations of $\mathfrak{g}$.

Proof:

A representation $\rho : G \rightarrow \text{Aut}(V)$ is a Lie group homomorphism, which by Lie's 2nd Theorem corresponds bijectively to a Lie algebra homomorphism $\mathfrak{g} \rightarrow \text{End}(V)$, i.e. a representation of $\mathfrak{g}$. ■

Chapter 5

Representation theory of $\mathfrak{su}(n)$

5.1 Representation theory of $\mathfrak{su}(2)$ and $\mathfrak{sl}(2)$

Remark.

Recall that $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ share the same Lie algebra (via the double cover $\mathrm{SU}(2) \twoheadrightarrow \mathrm{SO}(3)$):

$$\mathfrak{su}(2) = \{A \in \mathrm{Mat}(2, \mathbb{C}) \mid \mathrm{tr}(A) = 0, A^\dagger = -A\}.$$

The Pauli matrix

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is Hermitian, so $\sigma_x^\dagger = \sigma_x \neq -\sigma_x$, and hence $\sigma_x \notin \mathfrak{su}(2)$. The associated basis element of $\mathfrak{su}(2)$ is $X := i\sigma_x$, and one has $\sigma_x = -iX$.

Definition 5.1.1: Complexification of a Lie algebra

Let $\mathfrak{g}$ be a real Lie algebra. Its **complexification** is the complex vector space

$$\mathfrak{g}_{\mathbb{C}} := \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C},$$

equipped with the complex-linear extension of the Lie bracket, defined on pure tensors by

$$[X \otimes \lambda, Y \otimes \lambda'] := [X, Y] \otimes \lambda\lambda' \quad \forall X, Y \in \mathfrak{g}, \lambda, \lambda' \in \mathbb{C}.$$

Example.

The complexification of $\mathfrak{su}(2)$ is

$$\mathfrak{sl}(2, \mathbb{C}) = \{A \in \mathrm{Mat}(2, \mathbb{C}) \mid \mathrm{tr}(A) = 0\}.$$

Remark.

Let $\mathfrak{g}$ be a real Lie algebra. Complex representations of $\mathfrak{g}_{\mathbb{C}}$, i.e. complex-linear Lie algebra homomorphisms $\mathfrak{g}_{\mathbb{C}} \rightarrow \mathrm{End}(V)$, are in bijection with real Lie algebra homomorphisms $\mathfrak{g} \rightarrow \mathrm{End}(V)$. The bijection is given by restriction to $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$, with inverse given by complex-linear extension.

To study the representations of $\mathfrak{su}(2)$, we use this bijection and work instead with the complex representations of $\mathfrak{sl}(2, \mathbb{C})$. We choose the standard basis

$$\begin{aligned} H &:= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z = L_z, \\ E &:= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \frac{1}{2}(\sigma_x + i\sigma_y) = L_+, \\ F &:= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \frac{1}{2}(\sigma_x - i\sigma_y) = L_-. \end{aligned}$$

One verifies directly that the commutation relations are

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

Proposition 5.1.2

Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(V)$ be a complex representation, and let $v \in V$ be an eigenvector of $\rho(H)$ with eigenvalue $\lambda \in \mathbb{C}$:

$$\rho(H)v = \lambda v.$$

Then $\rho(E)v$ is either zero or an eigenvector of $\rho(H)$ with eigenvalue $\lambda + 2$, and $\rho(F)v$ is either zero or an eigenvector of $\rho(H)$ with eigenvalue $\lambda - 2$.

Proof:

Since ρ is a Lie algebra homomorphism,

$$\rho(H)\rho(E)v - \rho(E)\rho(H)v = \rho([H, E])v = \rho(2E)v = 2\rho(E)v.$$

Hence

$$\rho(H)\rho(E)v = \rho(E)\rho(H)v + 2\rho(E)v = \lambda\rho(E)v + 2\rho(E)v = (\lambda + 2)\rho(E)v.$$

The same computation using $[H, F] = -2F$ gives

$$\rho(H)\rho(F)v = (\lambda - 2)\rho(F)v.$$

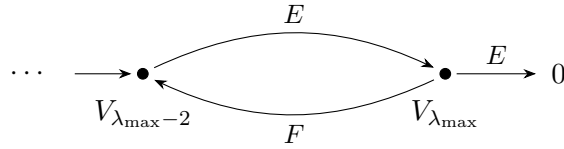
Thus $\rho(E)$ raises and $\rho(F)$ lowers the $\rho(H)$ -eigenvalue by 2. Denote by

$$V_\mu := \ker(\rho(H) - \mu \text{id}_V)$$

the μ -eigenspace of $\rho(H)$. Starting from a single eigenvector $v \in V_\lambda$, let V' be the subrepresentation generated by acting with $\rho(E)$ and $\rho(F)$ on v . Then

$$V' = \bigoplus_{n \in \mathbb{Z}} V_{\lambda+2n}.$$

If V is finite-dimensional, only finitely many summands $V_{\lambda+2n}$ are non-zero. In particular, there exists a maximal eigenvalue $\lambda_{\max}$ such that $\rho(E)$ annihilates $V_{\lambda_{\max}}$:



Definition 5.1.3: Highest weight vector

Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(V)$ be a complex representation. A non-zero vector $v \in V$ is called a **highest weight vector** if

$$\rho(E)v = 0.$$

Its $\rho(H)$ -eigenvalue is called the **highest weight** of v .

Proposition 5.1.4

Let $V = \bigoplus_{n \in \mathbb{Z}} V_{\lambda+2n}$ be an irreducible complex representation of $\mathfrak{sl}(2, \mathbb{C})$, and let $v \in V_{\alpha}$ with $\alpha = \lambda_{\max}$ be a highest weight vector. Then V is generated as a vector space by $\{\rho(F)^k v\}_{k \in \mathbb{N}_0}$.

Proof:

Let $W := \text{span}_{\mathbb{C}}\{\rho(F)^k v \mid k \in \mathbb{N}_0\} \subseteq V$. We show that W is a subrepresentation by checking invariance under each of $\rho(F)$, $\rho(H)$, and $\rho(E)$. Invariance under $\rho(F)$:

$$\rho(F) \cdot \rho(F)^k v = \rho(F)^{k+1} v \in W$$

Invariance under $\rho(H)$. Since $\rho(F)^k v \in V_{\alpha-2k}$

$$\rho(H)\rho(F)^k v = (\alpha - 2k)\rho(F)^k v \in W$$

Invariance under $\rho(E)$. Since $[E, F] = H$, we have

$$\rho(E)\rho(F) = \rho(F)\rho(E) + \rho(H)$$

For $k = 1$:

$$\rho(E)\rho(F)v = \rho(F)\rho(E)v + \rho(H)v = 0 + \alpha v = \alpha v.$$

For general k , apply this identity repeatedly to expand $\rho(E)\rho(F)^k v$:

$$\begin{aligned} \rho(E)\rho(F)^k v &= \rho(F)\rho(E)\rho(F)^{k-1} v + \rho(H)\rho(F)^{k-1} v \\ &= \rho(F)^2 \rho(E)\rho(F)^{k-2} v + \rho(F)\rho(H)\rho(F)^{k-2} v + (\alpha - 2k + 2)\rho(F)^{k-1} v \\ &= \dots \\ &= \rho(F)^k \rho(E)v + (\alpha + (\alpha - 2) + \dots + (\alpha - 2k + 2))\rho(F)^{k-1} v. \end{aligned}$$

Since $\rho(E)v = 0$ and the sum has k terms,

$$\rho(E)\rho(F)^k v = k(\alpha - k + 1)\rho(F)^{k-1} v \in W.$$

Thus W is a subrepresentation. Since $v \neq 0$ we have $W \neq \{0\}$, and since V is irreducible we

conclude $W = V$.

Remark.

The proof above also establishes the formula

$$\rho(E)\rho(F)^k v = k(\alpha - k + 1)\rho(F)^{k-1}v \quad \forall k \in \mathbb{N}_0.$$

Corollary 5.1.5

All weight spaces V_κ are 1-dimensional.

Proof:

The vectors $v, \rho(F)v, \dots, \rho(F)^\alpha v$ form a basis of V (as shown in the proof of the following corollary). Since $\rho(F)^k v \in V_{\alpha-2k}$ and the eigenvalues $\alpha, \alpha-2, \dots, -\alpha$ are pairwise distinct, each eigenspace is spanned by exactly one basis vector, hence 1-dimensional.

Corollary 5.1.6

The highest weight satisfies $\alpha \in \mathbb{N}_0$, and $\dim V = \alpha + 1$.

Proof:

Since V is finite-dimensional and the non-zero vectors among $\{\rho(F)^k v\}_{k \geq 0}$ are linearly independent (they lie in distinct eigenspaces $V_{\alpha-2k}$ of $\rho(H)$), there exists a smallest $k_1 \in \mathbb{N}$ with $\rho(F)^{k_1} v = 0$.

Since $\rho(F)^{k_1} v = 0$, we have $\rho(E)\rho(F)^{k_1} v = 0$. Applying the formula from the remark gives

$$0 = k_1(\alpha - k_1 + 1)\rho(F)^{k_1-1}v.$$

By minimality of k_1 , the vector $\rho(F)^{k_1-1}v \neq 0$. Since $k_1 \geq 1$, both k_1 and $\rho(F)^{k_1-1}v$ are non-zero, so we must have

$$\alpha - k_1 + 1 = 0 \quad \Rightarrow \quad \alpha = k_1 - 1 \in \mathbb{N}_0.$$

The vectors $v, \rho(F)v, \dots, \rho(F)^{k_1-1}v$ are linearly independent and span V by the previous proposition, so $\dim V = k_1 = \alpha + 1$.

Example.

- $\mathfrak{sl}(2, \mathbb{C})$ has a 1-dimensional representation ($\alpha = 0$):

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(\mathbb{C}), \quad X \mapsto 0.$$

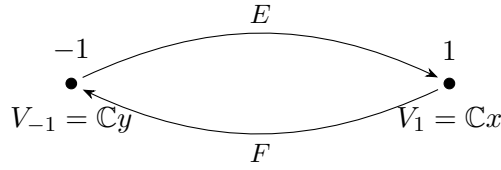
The single weight space is $V_0 = \mathbb{C}$:

$$\bullet \\ V_0$$

- $\mathfrak{sl}(2, \mathbb{C})$ has a 2-dimensional representation ($\alpha = 1$), the **standard representation**:

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(\mathbb{C}^2), \quad X \mapsto X.$$

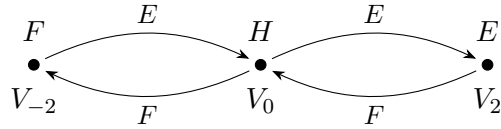
Let x, y be a basis of $\mathbb{C}^2$ with $\rho(H)x = x$ and $\rho(H)y = -y$. Then $V_1 = \mathbb{C}x$ and $V_{-1} = \mathbb{C}y$:



- $\mathfrak{sl}(2, \mathbb{C})$ acts on itself ($\alpha = 2$) by

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(\mathfrak{sl}(2, \mathbb{C})), \quad X \mapsto [X, \cdot].$$

The weight spaces are $V_{-2} = \mathbb{C}F$, $V_0 = \mathbb{C}H$, $V_2 = \mathbb{C}E$:



This is called the **adjoint representation**.

Fact 5.1.7

All finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$ representations are fully reducible.

Definition 5.1.8: Tensor product representation

Let V and W be representations of a Lie algebra $\mathfrak{g}$. The **tensor product representation** on $V \otimes W$ is defined by

$$\rho_{V \otimes W}(X) := \rho_V(X) \otimes \text{id}_W + \text{id}_V \otimes \rho_W(X) \quad \forall X \in \mathfrak{g}.$$

(Verification that this is a representation: see exercise sheets.)

Example.

Isospin (Heisenberg, 1932). When only the strong interaction is considered, there is an approximate symmetry exchanging proton and neutron. This makes $\mathbb{C}[p, n]$ a copy of the standard representation (highest weight $\alpha = 1$), with the proton at weight $+1$ and the neutron at weight -1 . Extending to the three light quarks,

$$\mathbb{C}[u, d, s] \cong V_1 \oplus V_0,$$

with u, d at weights ± 1 and s at weight 0 , where V_n denotes the irreducible representation with highest weight n . Mesons transform in the following multiplets:

- π^+, π^0, π^- form an isospin triplet in V_2 ;
- (K^+, K^0) and $(K^-, \bar{K}^0)$ form isospin doublets in V_1 .

Tensor product $V_1 \otimes V_1$. Weights add under tensor products: $\rho(H)(v \otimes w) = (\lambda_v + \lambda_w) v \otimes w$. For V_1 with basis $\{x, y\}$ at weights $+1, -1$:

$$\begin{array}{ccc}
y \otimes y & y \otimes x, & x \otimes y & x \otimes x \\
\bullet & \bullet \bullet & \bullet & \\
-2 & 0 & 2 &
\end{array}$$

The weight-0 space is 2-dimensional, so $V_1 \otimes V_1 \cong V_2 \oplus V_0$.

Tensor product $V_2 \otimes V_1$. Let $\{|\uparrow\rangle, |0\rangle, |\downarrow\rangle\}$ be a basis of V_2 at weights $+2, 0, -2$. The six tensor products and their weights are:

	$x (+1)$	$y (-1)$
$ \uparrow\rangle (+2)$	$ \uparrow\rangle \otimes x (+3)$	$ \uparrow\rangle \otimes y (+1)$
$ 0\rangle (0)$	$ 0\rangle \otimes x (+1)$	$ 0\rangle \otimes y (-1)$
$ \downarrow\rangle (-2)$	$ \downarrow\rangle \otimes x (-1)$	$ \downarrow\rangle \otimes y (-3)$

The weight spaces have dimensions 1, 2, 2, 1 at weights $+3, +1, -1, -3$, consistent with $V_2 \otimes V_1 \cong V_3 \oplus V_1$.

Theorem 5.1.9: Classification of irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$

The irreducible complex representations of $\mathfrak{sl}(2, \mathbb{C})$ are, up to isomorphism, exactly $\text{Sym}^n(\mathbb{C}^2)$ for $n \in \mathbb{N}_0$, where $\mathbb{C}^2$ carries the standard representation $X \mapsto X$. In particular, $\text{Sym}^n(\mathbb{C}^2)$ is $(n+1)$ -dimensional with highest weight n .

Proof:

Let x, y be the standard basis of $\mathbb{C}^2$. The space $\text{Sym}^n(\mathbb{C}^2)$ is $(n+1)$ -dimensional with basis $\{x^{n-k}y^k\}_{k=0}^n$. Using the Leibniz rule together with $\rho(H)x = x$ and $\rho(H)y = -y$:

$$\rho(H)(x^{n-k}y^k) = (n-k)(+1)x^{n-k}y^k + k(-1)x^{n-k}y^k = (n-2k)x^{n-k}y^k.$$

The highest weight is n , achieved by x^n . Since $\rho(E)x = 0$ in the standard representation, $\rho(E)(x^n) = 0$, so x^n is a highest weight vector. Using $\rho(F)x = y$ and $\rho(F)y = 0$, repeated application gives

$$\rho(F)^k(x^n) = \frac{n!}{(n-k)!} x^{n-k}y^k \neq 0 \quad \text{for } k = 0, 1, \dots, n.$$

By the previous proposition, $\text{Sym}^n(\mathbb{C}^2)$ is generated by x^n and is therefore irreducible.

Conversely, every finite-dimensional irreducible representation has highest weight $\alpha \in \mathbb{N}_0$ and dimension $\alpha + 1$ (by the corollary). Since $\text{Sym}^\alpha(\mathbb{C}^2)$ is irreducible of dimension $\alpha + 1$ with highest weight α , it is the unique such representation up to isomorphism. $\blacksquare$

5.1.1 Unitary representations and the physics convention

Remark.

In physics, one is interested in **unitary representations**, i.e. group homomorphisms $G \rightarrow U(n)$. At the Lie algebra level, this means maps into

$$\mathfrak{u}(n) = \{A \in M_n(\mathbb{C}) \mid A^\dagger = -A\},$$

so every generator acts as a skew-adjoint operator. This condition is not preserved by complexification ($\otimes_{\mathbb{R}} \mathbb{C}$): the complexification introduces new elements for which no natural adjointness condition is imposed.

Both $\mathfrak{su}(2)$ and $\mathfrak{sl}(2, \mathbb{R})$ are real forms of $\mathfrak{sl}(2, \mathbb{C})$, but their unitary representation theories are opposite in character:

- $\mathfrak{su}(2)$: all finite-dimensional representations can be made unitary (the associated Lie group $SU(2)$ is compact).
- $\mathfrak{sl}(2, \mathbb{R})$: there are no non-trivial finite-dimensional unitary representations.

The standard basis of $\mathfrak{su}(2)$, expressed in terms of H, E, F , is

$$Z = iH = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad Y = E - F = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X = i(E + F) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

One checks that each satisfies $A^\dagger = -A$. In any unitary $\mathfrak{su}(2)$ -representation, the skew-adjointness of $Z = iH$ forces $\rho(H)$ to be self-adjoint, while the skew-adjointness of Y and X together give

$$\rho(E)^\dagger = \rho(F).$$

Example.

The standard representation on $\mathbb{C}^2$ is unitary: $SU(2) \subset GL(2, \mathbb{C})$ acts by isometries of the standard Hermitian inner product. Since tensor products of unitary representations are again unitary, $\text{Sym}^n(\mathbb{C}^2)$ is unitary for all $n \in \mathbb{N}_0$.

Remark.

Convention. For a representation $\rho : \mathfrak{g} \rightarrow \text{End}(V)$ we write $Xv := \rho(X)(v)$ for $X \in \mathfrak{g}$ and $v \in V$, suppressing ρ from the notation.

The basis $\{F^k v\}_{k=0}^n$ of V_n is orthogonal (the vectors lie in distinct $\rho(H)$ -eigenspaces, and $\rho(H)$ is self-adjoint), but not orthonormal. Using $\rho(E)^\dagger = \rho(F)$, which gives $\langle u | Fw \rangle = \langle Eu | w \rangle$, together with $EF^k v = k(n - k + 1)F^{k-1}v$, one computes

$$\|F^k v\|^2 = \langle F^k v | F \cdot F^{k-1} v \rangle = \langle EF^k v | F^{k-1} v \rangle = k(n - k + 1) \|F^{k-1} v\|^2.$$

Starting from $\|v\|^2 = 1$, this recursion gives $\|F^k v\|^2 = k! \frac{n!}{(n-k)!}$ and defines a positive-definite inner product with respect to which the representation is unitary.

Physics basis. In quantum mechanics, states are labelled by eigenvalues of simultaneously measurable observables. Setting $j := n/2$ (total angular momentum quantum number), the orthonormal basis is defined inductively by

$$|j, j\rangle := v, \quad |j, m-1\rangle := \frac{F|j, m\rangle}{\sqrt{(j+m)(j-m+1)}}, \quad m = -j+1, \dots, j.$$

The second label m (magnetic quantum number, or z-projection of angular momentum) runs over $m = -j, -j+1, \dots, j$. In this basis the generators act by the standard angular

momentum formulas:

$$\begin{aligned} H |j, m\rangle &= 2m |j, m\rangle, \\ E |j, m\rangle &= \sqrt{(j-m)(j+m+1)} |j, m+1\rangle. \end{aligned}$$

5.2 Representation theory of $\mathfrak{su}(3)$ and $\mathfrak{sl}(3, \mathbb{C})$

Remark.

Physics background: the eightfold way. The three lightest quarks — up (u), down (d), and strange (s) — have masses

$$m_u \approx 2.2 \text{ MeV}, \quad m_d \approx 4.7 \text{ MeV}, \quad m_s \approx 93 \text{ MeV}.$$

All three are much lighter than the typical hadronic scale (~ 1 GeV), giving rise to an approximate $SU(3)$ -flavor symmetry acting on $\mathbb{C}[u, d, s]$. Historically, physicists observed that mesons and baryons organize into multiplets matching representations of $\mathfrak{sl}(3)$ — a phenomenon Gell-Mann called the **eightfold way**, named after the 8-dimensional adjoint representation in which the light mesons sit. Some of the predicted particles were initially absent from experiment; quarks were introduced as a mathematical bookkeeping device to explain the multiplet structure, and the missing particles were subsequently discovered. This led to Gell-Mann's Nobel Prize in 1969.

Definition 5.2.1: Cartan subalgebra of $\mathfrak{sl}(3, \mathbb{C})$

The **Cartan subalgebra** of $\mathfrak{sl}(3, \mathbb{C})$ is

$$\mathfrak{h} := \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \mid a + b + c = 0 \right\} \subseteq \mathfrak{sl}(3, \mathbb{C}).$$

Since diagonal matrices commute, $\mathfrak{h}$ is abelian: $[X, Y] = 0$ for all $X, Y \in \mathfrak{h}$. A basis is given by

$$H_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad H_{23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The role of $\mathfrak{h}$ is analogous to that of $\mathbb{C}H$ in $\mathfrak{sl}(2)$: it is the subalgebra whose simultaneous eigenvectors we use to organise representations.

Fact 5.2.2

In every finite-dimensional $\mathfrak{sl}(3, \mathbb{C})$ -representation, all elements of $\mathfrak{h}$ act by simultaneously diagonalizable operators.

Definition 5.2.3: Weight

Let V be a finite-dimensional $\mathfrak{sl}(3, \mathbb{C})$ -representation. A vector $v \in V$ is a **weight vector** with **weight** $\alpha \in \mathfrak{h}^*$ if

$$\rho(H)v = \alpha(H)v \quad \forall H \in \mathfrak{h}.$$

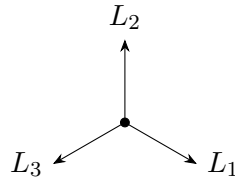
The corresponding **weight space** is $V_\alpha = \{v \in V \mid \rho(H)v = \alpha(H)v \forall H \in \mathfrak{h}\}$. By the fact above, $V = \bigoplus_\alpha V_\alpha$ (finite direct sum over $\mathfrak{h}^*$).

Remark.

Define linear functionals $L_1, L_2, L_3 \in \mathfrak{h}^*$ by

$$L_i \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} := a_i.$$

Since every $H \in \mathfrak{h}$ satisfies $a_1 + a_2 + a_3 = 0$, we have $L_1 + L_2 + L_3 = 0$, so $\{L_1, L_2\}$ is a basis of $\mathfrak{h}^*$ with $L_3 = -L_1 - L_2$. The three functionals are symmetric and point in directions 120° apart in $\mathfrak{h}^*$:

**Example.**

The defining 3-dimensional representation $\mathfrak{sl}(3, \mathbb{C}) \rightarrow \text{End}(\mathbb{C}^3)$, $X \mapsto X$, decomposes into weight spaces as

$$\mathbb{C}^3 = \mathbb{C}_{L_1} \oplus \mathbb{C}_{L_2} \oplus \mathbb{C}_{L_3},$$

where e_i has weight L_i : $He_i = a_i e_i = L_i(H)e_i$ for $H = \text{diag}(a_1, a_2, a_3)$.

Remark.

In $\mathfrak{sl}(2)$ we found eigenvectors of $\text{ad}(H)$ (i.e. $[H, E] = 2E$, $[H, F] = -2F$) to build a basis that moves between weight spaces. We do the same for $\mathfrak{sl}(3)$: decompose $\mathfrak{sl}(3, \mathbb{C})$ into weight spaces under the adjoint representation $X \mapsto [X, -]$.

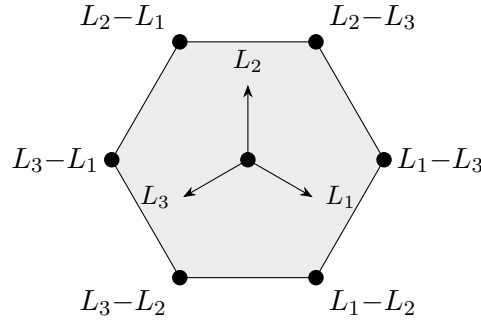
Since $[\mathfrak{h}, \mathfrak{h}] = 0$, the Cartan subalgebra sits at weight 0. The remaining basis elements are the **matrix units** E_{ij} (with 1 in position (i, j) and 0 elsewhere, $i \neq j$). For $H = \text{diag}(a_1, a_2, a_3)$:

$$\begin{aligned} [H, E_{12}] &= \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \\ &= \begin{pmatrix} 0 & a_1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & a_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = (a_1 - a_2) E_{12} = (L_1 - L_2)(H) E_{12}. \end{aligned}$$

The same computation gives $[H, E_{ij}] = (L_i - L_j)(H) E_{ij}$ for all $H \in \mathfrak{h}$, so E_{ij} is a weight vector with weight $L_i - L_j$. This gives the **root space decomposition**

$$\mathfrak{sl}(3, \mathbb{C}) = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{L_i - L_j}, \quad \mathfrak{g}_{L_i - L_j} = \mathbb{C} E_{ij}.$$

The six weights $L_i - L_j$ ($i \neq j$), called **roots**, form a regular hexagon in $\mathfrak{h}^*$. The total dimension $\dim \mathfrak{sl}(3, \mathbb{C}) = 2 + 6 = 8$ matches the eightfold way.



Proposition 5.2.4

Let $X, Y \in \mathfrak{sl}(3, \mathbb{C})$ be weight vectors of weights α and β . Then $[X, Y]$ has weight $\alpha + \beta$, i.e.

$$[X, Y] \in \mathfrak{g}_{\alpha + \beta},$$

where $\mathfrak{g}_\gamma := 0$ whenever γ is not a root.

Proof:

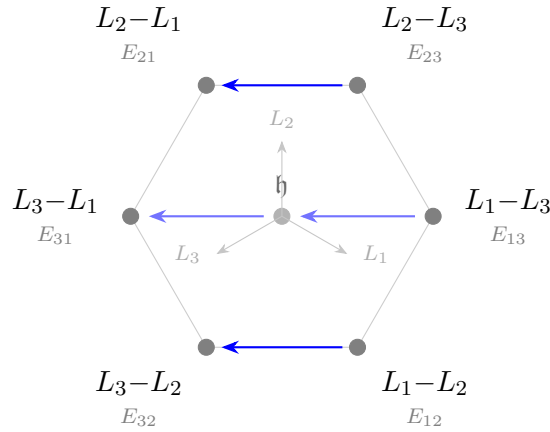
For any $H \in \mathfrak{h}$, the Jacobi identity gives

$$\begin{aligned} [H, [X, Y]] &= -[Y, [H, X]] - [X, [Y, H]] \\ &= -[Y, \alpha(H)X] + [X, \beta(H)Y] \\ &= (\alpha(H) + \beta(H)) [X, Y]. \end{aligned}$$

Since this holds for all $H \in \mathfrak{h}$, the element $[X, Y]$ is a weight vector of weight $\alpha + \beta$. ■

Example.

In the diagram, taking the commutator with E_{ij} moves every weight in the E_{ij} -direction. The diagram below shows the action of $[E_{31}, -]$, which shifts by $L_3 - L_1$, i.e. one step to the left. Moving outside the hexagon gives zero.



Proposition 5.2.5

Let V be a finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$, let $v \in V$ be a weight vector of weight α , and let $X \in \mathfrak{sl}(3, \mathbb{C})$ be of weight β . Then $X(v) \in V$ is of weight $\alpha + \beta$.

Proof:

For any $H \in \mathfrak{h}$,

$$H(X(v)) = X(H(v)) + [H, X](v) = (\alpha + \beta)(H) X(v).$$

Remark.

It follows that the weights appearing in an irreducible representation differ by integer combinations of the roots. The lattice generated by the roots,

$$\Lambda_R \subseteq \mathfrak{h}^*,$$

is called the **root lattice**.

There is no canonical notion of a highest weight, so we must make a choice.

Definition 5.2.6: Highest weight map

A **highest weight map** is a linear map $\ell: \mathfrak{h}^* \rightarrow \mathbb{C}$ such that

$$\operatorname{Re}(\ell(\alpha)) \neq 0 \quad \text{for all } 0 \neq \alpha \in \Lambda_R.$$

A root α is called **positive** if $\ell(\alpha) > 0$, and **negative** otherwise.

To be concrete, for $\mathfrak{sl}(3)$ we pick

$$\ell(a_1 L_1 + a_2 L_2 + a_3 L_3) = a a_1 + b a_2 + c a_3, \quad a + b + c = 0, \quad a > b > c.$$

This particular choice does not matter.

Definition 5.2.7: Highest weight vector ($\mathfrak{sl}(3)$)

Let V be a representation of $\mathfrak{sl}(3, \mathbb{C})$. A weight vector $v \in V_\alpha$ is called a **highest weight vector** if $\operatorname{Re} \ell(\alpha)$ is maximal among all weights of V , or equivalently if the positive root vectors annihilate it:

$$E_{12} v = E_{23} v = E_{13} v = 0.$$

Remark.

The word *weight* is used in several related senses, which are worth keeping apart:

- A **weight** is, abstractly, an element $\alpha \in \mathfrak{h}^*$ of the dual of the Cartan subalgebra — a linear functional on $\mathfrak{h}$.
- The weight of a **vector** v in a representation $\rho: \mathfrak{sl}(3, \mathbb{C}) \rightarrow \operatorname{End}(V)$ is the $\alpha \in \mathfrak{h}^*$ with

$$\rho(H) v = \alpha(H) v \quad \forall H \in \mathfrak{h};$$

such a v is a *weight vector* of weight α . (We usually omit ρ and write $Hv := \rho(H)v$.)

- The **weight space** V_α is the span of all weight vectors of weight α . A weight of the **representation** V is then an α with $V_\alpha \neq 0$, i.e. one that actually occurs in the decomposition $V = \bigoplus_\alpha V_\alpha$.
- In the special case of the **adjoint representation**, where $\mathfrak{sl}(3, \mathbb{C})$ acts on itself by $X \mapsto [X, -]$, the (non-zero) weights are exactly the **roots** $L_i - L_j$, and the corresponding weight vectors are the root vectors E_{ij} .

Proposition 5.2.8

Let V be an irreducible finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$ and $v \in V_\alpha$ a highest weight vector. Then V is generated by the elements obtained from v by successively applying E_{21} , E_{31} , and E_{32} .

Proof:

Since V is irreducible, it suffices to show that the subspace generated in this way is a non-zero subrepresentation; irreducibility then forces it to equal V .

Let $W \leq V$ be the subspace generated by acting on v with E_{21} , E_{31} , and E_{32} . We need to show that acting with the positive root vectors E_{12} , E_{13} , E_{23} also preserves W . Since $E_{13} = [E_{12}, E_{23}]$, it is enough to show $E_{12}(W) \subseteq W$ and $E_{23}(W) \subseteq W$.

Note that W can equally be generated by acting with only E_{21} and E_{32} , since $[E_{21}, E_{32}] = -E_{31}$.

We proceed by induction. Define W_n to be the subspace of W generated by applying words of length at most n in E_{21}, E_{32} to v , with $W_{-1} := 0$ and $W_0 := \mathbb{C}v$. We show that acting with a positive root vector maps W_n into W_{n-1} .

Let $w_n \in W_n$. There are two cases,

$$w_n = E_{21} w_{n-1} \quad \text{or} \quad w_n = E_{32} w_{n-1}, \quad w_{n-1} \in W_{n-1}.$$

In the first case,

$$\begin{aligned} E_{12} E_{21} w_{n-1} &= E_{21} E_{12} w_{n-1} + \underbrace{[E_{12}, E_{21}]}_{\in \mathfrak{h}} w_{n-1} \\ &= E_{21} \underbrace{E_{12} w_{n-1}}_{\in W_{n-2}} + \alpha([E_{12}, E_{21}]) w_{n-1}. \end{aligned}$$

By the induction hypothesis $E_{12} w_{n-1} \in W_{n-2}$, so the first term lies in W_{n-1} ; the second term lies in W_{n-1} as well. Hence $E_{12} w_n \in W_{n-1}$.

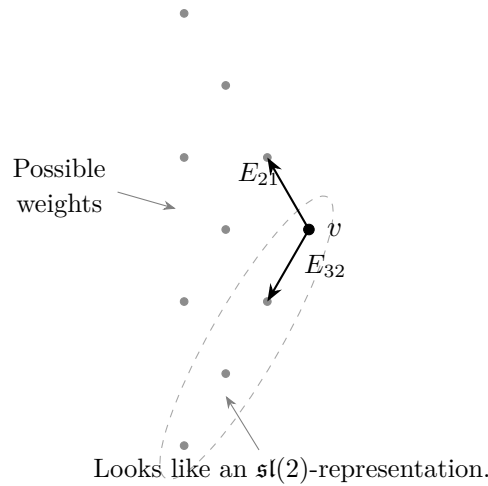
The remaining cases follow in the same way from

$$[E_{23}, E_{12}] = 0, \quad [E_{12}, E_{32}] = 0, \quad [E_{23}, E_{32}] \in \mathfrak{h}.$$

Remark.

Let V be a finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$ and $v \in V_\alpha$ a highest weight vector. Then acting with the negative roots on v generates an irreducible subrepresentation.

The weights reachable from v fill a cone opening in the directions of the negative roots. Along a single root string the picture reduces to a familiar $\mathfrak{sl}(2, \mathbb{C})$ -representation.



Definition 5.2.9: Lie subalgebra

Let $\mathfrak{g}$ be a Lie algebra. A **Lie subalgebra** is a subspace $\mathfrak{h} \leq \mathfrak{g}$ such that

$$[X, Y] \in \mathfrak{h} \quad \forall X, Y \in \mathfrak{h}.$$

Example.

For each pair $1 \leq i < j \leq 3$, the matrices E_{ij}, E_{ji}, H_{ij} span a Lie subalgebra of $\mathfrak{sl}(3, \mathbb{C})$ isomorphic to $\mathfrak{sl}(2, \mathbb{C})$, with E_{ij}, E_{ji}, H_{ij} taking the roles of E, F, H .

Consequently the possible highest weights are labelled by two non-negative integers $(n_{12}, n_{23}) \in$

$\mathbb{N}_0^2$, and the corresponding highest weight is

$$(n_{12} + n_{23}) L_1 + n_{23} L_2 = n_{12} L_1 - n_{23} L_3.$$

In particular, every weight occurring in a finite-dimensional representation is an integer combination $aL_1 + bL_2 + cL_3$ with $a, b, c \in \mathbb{Z}$.

Definition 5.2.10: Weight lattice

The **weight lattice** $\Lambda_W \subseteq \mathfrak{h}^*$ is the lattice generated by L_1, L_2, L_3 . It contains the root lattice:

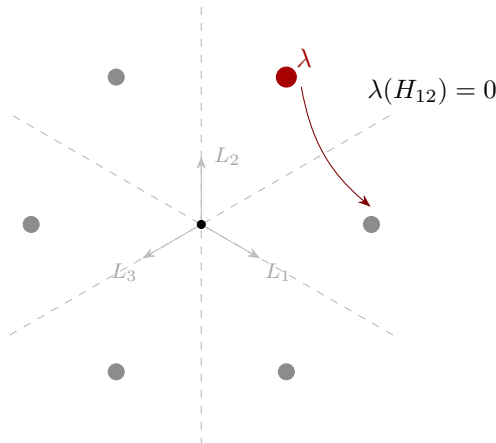
$$\Lambda_R \subseteq \Lambda_W \subseteq \mathfrak{h}^*.$$

Remark.

Restricting a representation to the $\mathfrak{sl}(2, \mathbb{C})$ -subalgebra generated by E_{ij}, E_{ji}, H_{ij} shows, exactly as in the $\mathfrak{sl}(2)$ theory, that its set of weights is symmetric under reflection in the hyperplane

$$\{\lambda \in \mathfrak{h}^* \mid \lambda(H_{ij}) = 0\}.$$

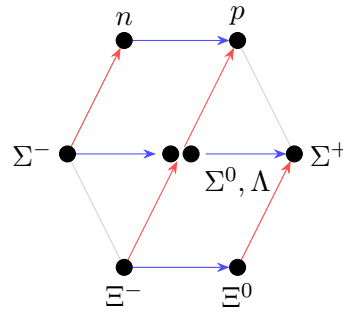
The group generated by these reflections is the **Weyl group**. Every weight diagram is invariant under it, so one completes a weight diagram by reflecting its weights in all the mirror hyperplanes.



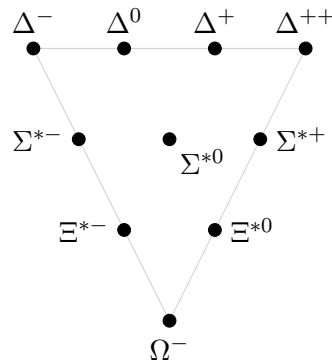
Recall the approximate $SU(3)$ -flavor symmetry rotating the light quarks u, d, s . One might ask why the relevant symmetry is the continuous group $SU(3)$ rather than a discrete group such as $\mathbb{Z}_3$; the answer is that the observed hadron multiplets match precisely the irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$, as the following examples illustrate.

Example.

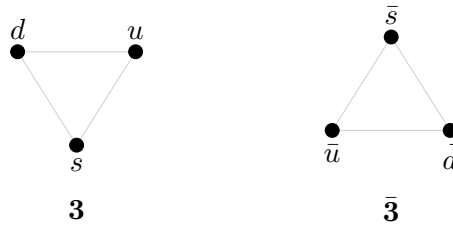
Baryon octet (adjoint representation, 8). The spin- $\frac{1}{2}$ baryons fill the weight diagram of the adjoint representation, a hexagon with a doubly-occupied centre:



Baryon decuplet (10). The spin- $\frac{3}{2}$ baryons fill a triangular weight diagram. The bottom vertex Ω^- was predicted before being discovered experimentally in 1964:



Quarks (3) and antiquarks ($\bar{3}$). The three quarks sit in the fundamental representation, the three antiquarks in its dual:



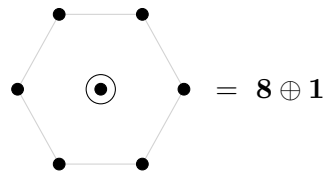
Tensor products. Recall that the tensor product representation acts by

$$\rho_{V \otimes W}(X) = \text{id}_V \otimes \rho_W(X) + \rho_V(X) \otimes \text{id}_W.$$

Combining a quark with an antiquark gives the meson multiplets,

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1},$$

a hexagon (the octet) together with a singlet at the centre:



Combining two quarks gives

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}},$$

where the symmetric sextet $\mathbf{6}$ consists of the diquark states

$$uu, \quad ud, \quad dd, \quad us, \quad ds, \quad ss.$$

These diquarks cannot exist as free particles in nature because of *confinement*.

Finally, $\mathbf{3} \otimes \mathbf{8} = \mathbf{15} \oplus \bar{\mathbf{6}} \oplus \mathbf{3}$ (left as an exercise).

We found that irreducible representations are determined by their highest weight. It remains to ask whether every allowed weight actually occurs as the highest weight of an irreducible representation. This can also be shown using Schur functors, but we take a different route.

Theorem 5.2.11: Classification of irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$

For each highest weight $\alpha = n_1 L_1 - n_2 L_3$ with $n_1, n_2 \in \mathbb{N}_0$, there exists a unique (up to isomorphism) irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight α .

Proof:

Existence. If V and W are representations with highest weights λ_V and λ_W , then $V \otimes W$ has highest weight $\lambda_V + \lambda_W$ (the tensor product of the two highest weight vectors is again a highest weight vector). Since $\mathbf{3}$ has highest weight L_1 and $\bar{\mathbf{3}}$ has highest weight $-L_3$, the representation

$$\mathbf{3}^{\otimes n_1} \otimes \bar{\mathbf{3}}^{\otimes n_2}$$

has highest weight $n_1 L_1 - n_2 L_3$. Picking a highest weight vector v in it, v generates an irreducible subrepresentation with highest weight $n_1 L_1 - n_2 L_3$. This proves existence.

Uniqueness. Let V and W be irreducible representations, both with highest weight α , and let $v \in V$, $w \in W$ be highest weight vectors. The representation $V \oplus W$ is not irreducible, but $v + w \in V \oplus W$ is again a highest weight vector of weight α . Let $\widetilde{W} \leq V \oplus W$ be the irreducible subrepresentation generated by $v + w$. The two projections

$$\widetilde{W} \rightarrow V, \quad \widetilde{W} \rightarrow W$$

are non-zero homomorphisms of representations, so by Schur's lemma they are isomorphisms. Hence

$$V \cong \widetilde{W} \cong W.$$

Remark.

(Added by transcriber, not part of the lecture.) Representations of a Lie group H classify the particles of any gauge theory with structure group H ; the following makes this precise for the Standard Model (*François*).

Gauge groups. A gauge theory with structure group H is built on a principal H -bundle $\pi: P \rightarrow M$ over spacetime M . The *gauge group*

$$\mathcal{H} = \text{Aut}_v(P) := \{ \Phi: P \xrightarrow{\sim} P \mid \pi \circ \Phi = \pi, \Phi(ph) = \Phi(p)h \}$$

is the group of fiber-preserving automorphisms; each element corresponds to an equivariant map $\gamma: P \rightarrow H$ satisfying $\gamma(ph) = h^{-1}\gamma(p)h$, and $\text{Lie } \mathcal{H} \cong \Omega_{\text{tens}}^0(P, \text{Lie } H)$. Incorporating spacetime diffeomorphisms gives the full symmetry group

$$\mathcal{G} = \text{Diff}(M) \ltimes \mathcal{H},$$

where $\text{Diff}(M)$ acts on $\mathcal{H}$ by conjugation via bundle lifts. Physical observables must be $\mathcal{H}$ -invariant and covariant under $\text{Diff}(M)$. For the Standard Model, $H = SU(3)_c \times SU(2)_L \times U(1)_Y$.

Connections and D^A . A *connection* $A \in \Omega_{\text{eq}}^1(P, \text{Lie } H)$ satisfies $\iota_{X\#} A = X$ for all $X \in \text{Lie } H$, splitting $TP = VP \oplus HP_A$ with horizontal projector $h_A: TP \rightarrow HP_A$. The *exterior covariant derivative* is the composition

$$D^A := \text{hor}_A \circ d : \Omega_{\text{tens}}^k(P, V) \xrightarrow{d} \Omega^{k+1}(P, V) \xrightarrow{\text{hor}_A} \Omega_{\text{tens}}^{k+1}(P, V),$$

where $(\text{hor}_A \alpha)(X_1, \dots, X_{k+1}) := \alpha(h_A X_1, \dots, h_A X_{k+1})$. On tensorial 0-forms, equivariance $(d\psi(X^\#) = -\rho_*(X)\psi)$ gives $D^A \psi = d\psi + \rho_*(A)\psi$. The *curvature*

$$F = dA + \frac{1}{2}[A, A] \in \Omega_{\text{tens}}^2(P, \text{Lie } H), \quad D^A F = 0 \quad (\text{Bianchi}),$$

transforms in the adjoint representation of H under $\mathcal{H}$:

$$A \mapsto \gamma^{-1} A \gamma + \gamma^{-1} d\gamma, \quad F \mapsto \gamma^{-1} F \gamma.$$

Gauge bosons. Each generator of $\text{Lie } H$ gives one gauge boson. The decomposition $\text{Lie } H = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ yields $8 + 3 + 1 = 12$: 8 gluons ($\mathfrak{su}(3)_c$), $W^\pm, W^3$ ($\mathfrak{su}(2)_L$), and B ($\mathfrak{u}(1)_Y$). The Yang–Mills Lagrangian $L_{\text{YM}} = \frac{1}{2} \text{Tr}(F \wedge *F)$ is $\mathcal{H}$ -invariant; a mass term $m^2 \text{Tr}(A \wedge *A)$ is not, since A transforms affinely—gauge invariance forces all gauge bosons massless.

Matter fields. A representation $\rho: H \rightarrow GL(V)$ specifies a matter field: a tensorial 0-form $\psi \in \Omega_{\text{tens}}^0(P, V) \cong \Gamma(P \times_H V)$. The *gauge charge* of ψ is the representation ρ ; minimal coupling to A is automatic via D^A , and $\langle \psi, \not{D}^A \psi \rangle$ is $\mathcal{H}$ -invariant by equivariance. For the quarks of each generation:

$$Q_L \sim (\mathbf{3}, \mathbf{2}, +\frac{1}{6}), \quad u_R \sim (\mathbf{3}, \mathbf{1}, +\frac{2}{3}), \quad d_R \sim (\mathbf{3}, \mathbf{1}, -\frac{1}{3}).$$

The representation content is experimental input, constrained by *anomaly cancellation*: certain traces over the fermion representations must vanish.

Mass generation. The Higgs field $\phi \in \Omega_{\text{tens}}^0(P, \mathbb{C}^2)$ in representation $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ of H couples to A via

$$L_{\text{Higgs}} = \langle D^A \phi, *D^A \phi \rangle - V(\phi), \quad V(\phi) = \mu^2 |\phi|^2 + \lambda |\phi|^4.$$

The full Lagrangian $L_{\text{YM}} + L_{\text{Higgs}} + \sum_\psi \langle \psi, \not{D}^A \psi \rangle + L_{\text{Yukawa}}$ is $\mathcal{H}$ -invariant and entirely determined by H and the representations $\{\rho_i\}$; all couplings follow from D^A , and L_{Yukawa} generates fermion masses. Gauge boson mass generation admits two equivalent descriptions.

Spontaneous symmetry breaking. For $\mu^2 < 0$, a nonzero vacuum ϕ_0 has stabilizer $U(1)_{\text{EM}} \subset H$; the residual gauge group is $SU(3)_c \times U(1)_{\text{EM}}$. The $W^\pm$ and Z acquire masses from $|D^A \phi_0|^2$, with the Goldstone modes absorbed as longitudinal polarizations; the photon remains massless.

Dressing field method (AttardFrançoisLazzariniMasson). A **dressing field** for a subgroup $K \subset H$ is a K -valued field $u: P \rightarrow K$ transforming as $u \mapsto \kappa^{-1}u$ under $\kappa \in K \subset \mathcal{H}$ (not as a connection). The composite fields

$$A^u := u^{-1}Au + u^{-1}du, \quad \psi^u := u^{-1}\psi$$

are K -invariant by construction. For $K = SU(2)_L$, the polar decomposition $\phi = u[\phi]|\phi|$ provides $u[\phi]$ as a dressing field; the $W^\pm, Z$ masses appear directly from $|D^A \phi|^2$ rewritten in terms of A^u , with $|\phi|$ the physical neutral Higgs. No vacuum, no Goldstone bosons, and no broken symmetry are invoked.

Dressing is not gauge fixing. Gauge fixing imposes a constraint $\chi(A) = 0$, selects one representative per $\mathcal{H}$ -orbit, and requires Faddeev-Popov ghosts. Dressing imposes no constraint: $(A, \phi) \mapsto (A^u, \psi^u)$ is a change of variables on the full field space, globally well-defined (no Gribov copies), with all degrees of freedom dynamical. Both pictures yield the same S-matrix observables; Elitzur's theorem—local gauge symmetry cannot be spontaneously broken—is automatically respected.

Flavor vs. gauge symmetry. The $SU(3)_c$ factor is an exact gauge symmetry: color is a conserved Noether charge and gluons couple identically to all quark flavors. The $SU(3)_{\text{flavor}}$ symmetry studied in this chapter is not a gauge symmetry: it is an accidental near-symmetry of the light-quark mass terms, acts on flavor indices, has no associated connection or gauge bosons, and is explicitly broken by quark mass differences.

Chapter 6

Simple Lie algebras

a

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